

Do floods induce long-term livelihood diversification? Evidence from Nigeria

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Abstract

Floods are among the most damaging climate-related shocks in low- and middle-income countries, but it remains unclear whether severe flood exposure induces households to diversify away from agriculture over the long term. This paper examines how flooding affects household livelihoods, labor allocation, and consumption in Nigeria using a nationally representative household panel spanning 2010–2019. We estimate impacts in a difference-in-differences framework that controls for predicted flood exposure and compare community-level flooding measured using survey reports of harmful flooding and a widely used database of satellite-detected inundation. Using survey-based flood exposure, we find persistent declines in consumption (12%) and agricultural production (17%) with no increase in non-farm enterprise labor or earnings. Wage employment rises only among households with prior wage work and without corresponding increases in wage earnings, while migration for work only increases after a delay of several years. In contrast, the satellite-based measure, constrained by low spatial resolution and high cloud cover during the flood period, identifies a substantially different set of flooded communities and suggests that flood exposure increases non-farm enterprise and total household earnings. These results imply that harmful flood exposure does not induce broad livelihood diversification in this setting, but that commonly-used flood exposure measures can lead to sharply different conclusions about climate adaptation responses.

JEL codes: J22, O13, Q10, Q54

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1 Introduction

Climate change is predicted to reduce agricultural productivity in many parts of the world due to increases in temperatures and in precipitation variability, which are compounded by damages from increasingly severe and frequent extreme weather events. A central question is whether such shocks induce households in low- and middle-income countries (LMICs) to diversify away from agriculture or instead reinforce existing livelihood patterns. The scope for on-farm adaptation is limited, and households may face substantial constraints to adjusting labor allocation and livelihood strategies in response to climate shocks. A large literature finds short-term migration or non-farm labor responses to weather shocks in LMICs, but evidence on long-term livelihood responses remains limited and mixed. This paper analyzes the long-term livelihood impacts of flood shocks by studying the effects of a major flooding event in Nigeria on household livelihood activities up to seven years afterward.

Floods are a particularly important source of risk and damages, accounting for 44% of all disasters globally and 31% of all economic losses from 1970-2019 (World Meteorological Association et al., 2021), and the largest projected increases in flood risk are in LMICs (Caretta et al., 2022). Floods can push households into new activities as coping strategies, but they may also decrease demand for non-farm goods and services and deplete assets needed to engage in non-farm activities. Consequently, it is unclear whether such shocks lead to persistent reallocation toward non-agricultural activities. We study these responses empirically in the context of widespread floods in Nigeria in 2012 that caused \$16.9 billion in damages and killed 431 people (Unah, 2021), using four rounds of the nationally representative General Household Survey-Panel (GHSP) from 2010-2019.

We analyze the impacts of community-level flood exposure using a difference-in-differences design comparing changes before and after the major flood events of 2012 for households that did and did not reside in communities exposed to those floods. This strategy generates ‘intent to treat’ estimates as most households in flooded communities do not report being directly affected. To account for potential endogeneity in community-level flood exposure, we balance the sample on recent flood history and underlying fluvial flood hazard, control for time-invariant correlates through household fixed effects, and recenter binary 2012 flood exposure around predicted exposure based on community geographical and socioeconomic characteristics. This recentering largely eliminates baseline differences in characteristics of communities and households by 2012 flood exposure.

Analyzing flood impacts requires identifying which households and communities were actually exposed to harmful flooding, but measuring flood exposure is challenging and different approaches may capture very different types of flooding and affected populations. Existing work on the economic impacts of floods uses a variety of data sources including administrative data, satellite imagery or radar, survey or media reports, or some combination of these sources. Each source has different limitations which can lead to inconsistency in identifying flood exposure (Bangalore et al., 2026; Guiteras et al., 2015; Patel, 2024; Saunders et al., 2025).¹ We consider the implications of us-

¹See Kar et al., 2024; Li et al., 2018; Tellman et al., 2021; Yan et al., 2015; Zhang et al., 2023 for examples of papers focusing on the question of flood detection.

ing a measure of community flood exposure based on GHSP survey reports versus one based on 250 m MODIS satellite imagery from NASA’s Global Flood Product (NASA, 2024). These measures identify substantially different sets of flooded communities, due to differences in the definition of ‘flooding’ and measurement issues. Survey reports identify when floods adversely affected community members: the subset of inundation events with realized harm. The NASA product provides a measure of physical inundation, but this also represents a subset of all inundation events. In particular, this satellite measure is less likely to identify floods in rural and agricultural communities due to resolution and cloud cover constraints.

Using survey-reported community flood exposure, we find persistent declines in household consumption and agricultural production following Nigeria’s 2012 floods, but no evidence of sustained livelihood diversification away from agriculture. However, these conclusions depend importantly on how flood exposure is measured. Satellite-detected floods decrease the average value of household farm production by 20% over the following survey waves but increase earnings from household non-farm enterprises (NFEs) by 45%, leading to an increase in total household earnings. The decrease in farm production is not significant in 2012-13, when floods would have directly affected agricultural activities, and there are no other adverse effects of flood exposure, consistent with the satellite measure not detecting a large share of survey-reported flooding. In contrast, survey-reported flooding in a community decreases household consumption per capita by 12% and farm production value by 17% on average, with the largest effects in the 2012-13 survey round and some evidence of persistence over the following two survey rounds. We also find a persistent 17% increase in the count of household members engaged in wage employment, but no changes in earnings from non-farm livelihood activities.

Focusing on the survey-reported measure, we find that estimated effects are not driven by differences in vulnerability to floods, differential attrition, or choices of controls and sample restrictions. The estimated effects of survey-reported flood exposure are economically large, particularly since just 24% of households in exposed communities report being directly affected. Decreased farm production value is driven by effects on crop production. Physical crop yields fall, especially in 2012-13, but we also find some evidence that households exposed to floods persistently reduce paid crop production inputs. Despite these effects we find no evidence of agricultural exit or reallocation of labor away from agriculture following flood exposure.

Nigerian households in the survey sample do not report commonly using migration or NFEs to cope with flood shocks. The probability that any household member migrates does not change on average, though we do find a significant 6 percentage point increase seven years after the floods, potentially suggesting short-term migration constraints. We find no average effects of flood exposure on household NFE engagement, labor supply, or earnings, with the exception of an increase in total labor hours supplied by household members in the post-harvest survey period. The results for wage employment are more nuanced. Flood exposure persistently increases the count of household members working for a wage and their total wage work hours, but these effects are concentrated among households engaged in wage work before the floods. Wage earnings do not increase despite

greater wage labor and we find suggestive evidence of lower daily wages earned by household members, which would be consistent with a combination of higher labor supply and lower labor demand following the floods. In line with these results, we find no evidence of changes in the household earning shares from different livelihood activities, which remain highly concentrated.

This paper contributes to our understanding of the dynamic impacts of weather-related disasters on labor supply and livelihoods in LMICs. Many studies examine the effects of floods on household outcomes, but evidence on migration and labor supply has focused primarily on South Asia and on short-term responses (see Section 2). Evidence from sub-Saharan Africa and on long-term livelihood effects remains limited. Using survey-reported measures of community flood exposure to a major flood event in Nigeria, we find persistent declines in household consumption per capita and farm production value. We find no increase in non-farm enterprise activity and that wage employment increases only among households already engaged in wage work, while migration rises only several years after the shock. These findings align with some prior evidence from South Asia (e.g., Sajid, 2023) but contrast with others (Mueller and Quisumbing, 2011; Patel, 2024), suggesting that responses to floods depend on context and shock characteristics and that non-farm livelihood responses may remain constrained for many households (see also Cinque et al., 2025).

The findings also speak to broader questions about structural transformation in LMICs (see e.g., Alobo Loison, 2015; Gollin and Kaboski, 2023) by examining whether extreme weather shocks induce reallocation of labor across activities. A growing literature analyzes how climate change – with a focus on temperatures – may affect livelihoods, and many studies evaluate short-term labor responses to weather shocks, but there is limited evidence on whether such shocks lead to long-term livelihood diversification or structural change in sub-Saharan Africa (Barrett et al., 2021). We find no evidence that exposure to Nigeria’s 2012 floods induces reallocation of labor from farm to non-farm activities. Household livelihoods remain concentrated following flood exposure, suggesting that severe flood shocks may not induce sustained diversification into non-farm activities in this setting. This has important implications in a context of increasing flood risk.

Finally, we show that a commonly used satellite-based measure of flooding identifies substantially different sets of communities than survey flood reports and lead to sharply different conclusions about the impacts of flooding. These results add to a growing literature on measurement concerns in evaluating the impacts of flood exposure (Bangalore et al., 2026; Patel, 2024; Saunders et al., 2025), and to a broader literature on using remote sensing in economic analysis and particularly the limitations of such measures.² Empirical studies of climate adaptation increasingly rely on remote sensing, but we show that remote-sensed physical inundation may diverge from survey-reported economically harmful exposure due both to differences in what each measure captures and to limitations in remote sensing capabilities. The striking differences we find in estimated impacts across measures highlight the need for greater attention to measurement, transparency, and sensitivity analysis in empirical work on floods.

²See Aker et al., 2026; Alix-Garcia and Millimet, 2023; Donaldson and Storeygard, 2016; Fowlie et al., 2019; Gibson et al., 2021; Jain, 2020; Josephson et al., 2026.

2 Conceptual framework and literature

The structural transformation of economies is a key aspect of modern economic development. Historically, structural transformation has involved decreasing employment in agriculture, hump-shaped employment shares in manufacturing, and increasing employment in services as a function of the level of economic development, though not all countries follow this trajectory (Sen, 2019). Models of structural transformation emphasize the role of technological progress in driving reallocation across sectors (Herrendorf et al., 2014). In agriculture, innovation-driven productivity growth tends to *push* labor out, as decreasing amounts of labor are needed to meet finite demand for agricultural products. In manufacturing and services, productivity growth *pulls* labor in by increasing the marginal product of labor and therefore wages while lower prices and increased incomes drive increasing demand for manufactured goods and services.

A large literature studies the drivers and constraints to structural transformation (see Gollin and Kaboski (2023) for a recent overview), but it is clear that differences across contexts in productivity across sectors can affect whether and how structural change occurs. These considerations make it important to consider how climate-related productivity shocks could affect structural transformation, particularly in LMICs with high shares of employment still engaged in agriculture.³ Climate change is decreasing agricultural productivity in most regions of the world. Recent research suggests that current adaptive measures are insufficient to mitigate these effects (Hultgren et al., 2025; Kala et al., 2023; Pörtner et al., 2022) and that constraints to migration and non-farm labor opportunities (Albert et al., 2021; Alobo Loison, 2015; Cruz and Rossi-Hansberg, 2024; Desmet and Rossi-Hansberg, 2024; Ibáñez et al., 2026) may increase the losses from climate change.

Several studies explore the impacts of climate change on structural transformation. Two global analyses find that decreased agricultural productivity due to rising temperatures increases agriculture’s share of employment as consumption expenditure falls and shifts towards subsistence goods (Cruz, 2024; Nath, 2025). At the national and sub-national level, there is evidence of both increases (Albert et al., 2021; Colmer, 2021; Pham, 2025) and decreases (Liu et al., 2023; Pham, 2025) in non-agricultural employment. What is clear from the literature is that reallocating labor to a more productive non-farm activity would reduce losses from permanent or transitory decreases in agricultural productivity.

In addition to increasing temperatures, climate change is also affecting the frequency and severity of extreme weather events. Many studies find that agricultural households respond to adverse weather events by increasing labor supply to non-agricultural activities in the same year in order to offset lost agricultural income and smooth household consumption. A growing body of evidence evaluates whether exposure to such shocks has persistent effects on household livelihood strategies,⁴ but the results are mixed so the implications for structural transformation are unclear.

³See Auffhammer and Kahn (2018) and Kala et al. (2023) for discussions of potential responses for agricultural households in LMICs.

⁴E.g., Mueller and Osgood (2009a, 2009b) on drought in Brazil, Efobi (2022) on drought in Nigeria, Alfani et al. (2024) on drought in Morocco, Kirchberger (2017) on earthquakes in Indonesia, and Van den Berg (2010) on hurricanes in Nicaragua. We discuss studies on the impacts of floods in more detail below.

If structural transformation is most commonly the result of productivity *increases* in the agricultural and non-agricultural sectors, how might an agricultural productivity *decrease* from a weather-related shock such as a flood promote labor reallocation? Several mechanisms may push households to exit agriculture or diversify their livelihood strategies. First, households may update their beliefs about risks and returns to agricultural production, though only if the shock is seen as an indicator of a changing risk profile and not as an extreme realization from an unchanged risk distribution. Second, the short-term non-farm labor and migration coping strategies documented in many studies to smooth consumption needs following shocks may help households learn about non-farm livelihood opportunities that are more productive than agriculture. Third, for weather disasters causing physical damages such as floods or cyclones, damage or loss of productive assets may change perceptions about sunk costs of remaining in agriculture. Forced displacement may similarly decrease the opportunity cost of allocating labor away from agriculture.

On the other hand, a variety of factors can constrain households wanting to respond to a shock by diversifying their livelihood activities or exiting agriculture. In developing contexts which often have high trade costs, low agricultural productivity can create a ‘food problem’ increasing the need to allocate land and labor to produce food and meet subsistence needs (Gollin et al., 2007; Nath, 2025). As documented in studies on climate change and structural transformation, reduced agricultural productivity also decreases local incomes. This will on average reduce demand for non-agricultural goods and services and associated non-agricultural labor (Liu et al., 2023), limiting local non-farm opportunities which may already be low in rural agricultural areas with weak infrastructure (Barrett et al., 2021). Migration is a possible response, but resource constraints following a productivity shock can interact with other barriers to prevent migration in search of non-agricultural work. Even successful migrants may face high costs of living and lower wages in urban areas due to increased market competition (Auffhammer and Kahn, 2018), and many rural households and individuals prefer not to migrate away permanently (Lagakos et al., 2023). Even where there are non-farm opportunities, household strategies to smooth consumption after a shock can deplete household assets (Carter et al., 2007), reducing resources available to invest in new skills, activities, and technologies that could be needed to engage in new livelihood activities.

This conceptual framework highlights how adverse shocks to agricultural productivity could have different effects on household livelihood strategies depending on the context and household characteristics. In well-integrated areas with more developed labor markets, agricultural households with sufficient resources may reallocate labor to non-farm activities while in less-integrated areas and for poorer households, reduced incomes and limited alternatives may increase engagement in agriculture. For example, Mueller et al. (2020) find that temperature and rainfall anomalies increase short-term out-migration in some African countries but decrease it in others and point to differences in non-farm opportunities and the need for subsistence production labor as potential mechanisms. Pham (2025) finds that extreme heat in Vietnam only leads to reallocation of labor away from agriculture in more globally-integrated regions.

Most weather-related shocks do not only affect agricultural productivity. This paper focuses

on floods, which can destroy or damage both agricultural and non-agricultural property and assets as well as local infrastructure. These other effects may affect livelihood responses to the shock. Local destruction can decrease non-agricultural work opportunities by disrupting infrastructure, supply chains, and demand, though reconstruction may increase demand at least temporarily. Livelihood effects of a flood or other weather shock may therefore differ from those of a pure shock to agricultural productivity that does not have broader economic effects (e.g., a crop disease). However, analyzing the livelihood effects of floods in an LMIC through the lens of agriculture remains relevant its continued importance for livelihoods and its vulnerability to weather shocks.

The evidence on the impacts of flood exposure on livelihoods, labor, and migration in LMICs is mixed and is concentrated in South Asian contexts, with relatively limited evidence from sub-Saharan Africa.⁵ For example, four studies report increases in short-term out-migration following flood exposure (Balboni et al., 2023; Berlemann and Tran, 2020; Patel, 2024; Pavel et al., 2023), two report decreases (Chen and Mueller, 2019; Chen et al., 2017), two report null effects (Gray and Mueller, 2012; Mueller et al., 2014), and one reports mixed results (Maystadt et al., 2016). While many studies find that flood exposure increases short-term non-farm labor activity (e.g., Maystadt et al., 2016; Mueller and Quisumbing, 2011; Vitellozzi and Giannelli, 2023), the small literature on long-term effects of floods on livelihood activities is less consistent and is confined to South Asian settings. Mueller and Quisumbing (2011) find that wages in both farm and non-farm activities fall and that workers move into non-farm casual labor in the short-term following flood exposure in Bangladesh, but that these effects do not persist 5 years later. Sajid (2023) finds that flood exposure in the past decade in rural areas of India decreases household wealth, non-agricultural work, and migration to urban areas while increasing engagement in agriculture. In contrast, Patel (2024) finds that floods in Bangladesh persistently increase migration and push employment out of agriculture, with effects mitigated by prior adaptation to past flooding exposure.

The different results across these studies motivate further evaluation of the long-term household labor supply and livelihood impacts of flood exposure in a new context (without the monsoon-related flooding prevalent in South Asia), and a need to identify factors which could lead to heterogeneous effects.

3 Context, data, and empirical approach

Floods are one of the most common types of natural disasters globally and the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report finds that there is high confidence that climate change is increasing the risk and severity of flooding, due largely to increases in extreme precipitation events (Caretta et al., 2022). Tellman et al. (2021) estimate that the proportion of the global population exposed to floods has increased by 20-24 percent from 2000 to 2015, and find that this will increase further by 2030 under current climate change projections. The largest projected increases are concentrated in South and Southeast Asia and in Sub-Saharan Africa (Caretta et al.,

⁵Studies in other contexts include Arnall et al., 2013 on resettlement and livelihoods in Mozambique and Berlemann and Tran, 2020 on migration in Vietnam.

2022). Rogers et al. (2025) estimate that by 2100, 63% of global flood exposure will be in the areas with the lowest GDP levels.

Nigeria is one of the countries most affected by increasing flood exposure. Tellman et al. (2021) estimate that the population exposed to flooding nearly doubled from 2000 to 2015. Rogers et al. (2025) project that it will nearly double again from 2020 to 2050, from 34 to 64 million at risk of 100-year floods of at least 10 cm. The Nigeria Hydrological Services Agency (NIHSA) reports that the frequency and severity of flooding have increased in recent years, disrupting agricultural production and livelihoods but also causing death, population displacement, and destruction of housing and other infrastructure (NIHSA 2021). In 2012, the most severe floods in Nigeria’s recorded history affected 30 of 36 states between September and November as a result of excessive rainfall and a major release of water from Lagdo Dam in neighboring Cameroon. The floods caused an estimated \$16.9 billion in damages, displaced over 2.1 million people, and killed 431 people (Unah, 2021).

Increasing flood risk and a recent catastrophic national flood disaster make Nigeria a relevant context for studying the economic impacts of flood exposure. Nigeria is also an important setting to study labor reallocation and structural transformation, with the largest population and economy in Africa. While 76% percent of Nigeria’s GDP comes from non-agricultural sectors, around 70% of Nigeria’s population is engaged in agriculture (Statista, 2023), primarily at low productivity for subsistence purposes (FAO, 2022).

3.1 Survey data

The main data source for the analysis is Nigeria’s General Household Survey Panel (GHSP).⁶ The GHSP, part of the World Bank’s Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA), is a nationally-representative panel survey including approximately 5,000 households conducted by the Nigeria National Bureau of Statistics. In each round, households are surveyed twice: once post-planting (around August-September) and once post-harvest (around February-March). Four survey rounds were been completed between 2010 and 2019, with two-thirds of the sample being replaced for the 2018-19 round. The sample of communities retained in the GHSP in 2018-19 was randomly selected. The analysis sample excludes all new households surveyed for the first time in 2018-19. Figure A1 shows a timeline of GHSP data collection. Efforts to track households even if they move result in quite low levels of attrition: just 7.7% of households drop out of the sample between 2010 and 2016.⁷ These efforts also allow us to track household-level migration over time.

The household survey includes detailed modules on household agricultural production, labor supply, assets, consumption, shocks, and socio-demographics. The primary household-level outcomes are measures of total annual earnings or production value by activity and engagement of members in different types of work. We consider three broad livelihood activities: household farm (crop and livestock) production, household non-farm enterprise (NFE), and wage employment. In

⁶See Online Appendix C for additional details.

⁷For the panel sample of the 2018-19 survey round, 95% of targeted households were tracked and included.

addition, we consider effects of flood exposure on measures of household consumption, food security, and assets, and on a variety of crop production outcomes.⁸ We convert all monetary values to constant 2016 PPP USD, accounting for inflation and changes in purchasing power, and winsorize continuous variables at the 99th percentile.

Earnings and production value are measured at the annual level over the 12 months prior to the post-harvest survey. We assign a value of 0 for households not engaged in particular activities. In 2010, 40% of households reported at least one member engaged in wage employment at some point in the year. Household wage earnings is the sum of individual members' earnings for all wage employment. Similarly, earnings from household non-farm enterprise (76% of households) is the sum of revenues from all enterprises – we do not subtract expenses or the value of household labor. For household farm activities (69% of households), we estimate the value of all crop and livestock production, again without subtracting expenses or labor costs. The value of crop production is based on crop sales revenue plus the value of unsold production, calculated as the quantity not sold times the median price for sales of that crop at the most precise of the household, community, state, or national level depending on whether there are any households selling the crop. We follow the same approach to value livestock products (e.g., animal labor, eggs, milk, meat, live animals). We also calculate total earnings from other sources, such as investments, remittances, and pensions.

We measure migration using information on household-level moves and changes in composition of household members. Household-level migration is defined by whether the household has moved to a different community between survey rounds. These movers remain in the survey sample, and 9.7% of the initial 2010-11 sample of households moved at some point over the following survey rounds. This measure will not capture temporary displacement if the household returns to its original community by the time of the next survey. Individual-level migration is defined using changes in household rosters. In each round, respondents are asked whether any previous members have left the household and whether any new members have joined. We construct measures of general member departures (for reasons other than death), and of member migration for work or better land and for marriage or other family-related reasons. We also measure whether any new adult member joined the household.

Each household is associated with community-level data based on surveys with a group of community informants conducted at the same time as the household surveys.⁹ The public survey data also include spatial characteristics calculated by the World Bank at the community level such as distances to markets and administrative centers, local climate and weather, and land cover. We complement this with data on fluvial flood hazard at a 90 meter resolution from the European Commission Joint Research Centre (JRC; Baugh et al., 2024) and data on rivers and bodies of water from the Digital Chart of the World. We calculate the distance between each survey community and the nearest body of water and the mean and maximum depth of 100-year floods within a 5 km radius as a measure of local flood hazard.

⁸Online Appendix C includes additional detail on the construction of all outcome variables.

⁹Communities are referred to as 'enumeration areas' (EAs) in the GHSP.

3.2 Flood data

An important challenge in analyzing floods is determining how to measure flood exposure – where floods have occurred. The main raw data sources include government records, media reports, survey reports, precipitation data, river or other water gauges, and satellite imagery or radar. All of these sources have been used in applied research studying the impacts of floods. Each source has different advantages and limitations.¹⁰ The most common flood incidence measures in recent years draw on high-frequency and high-resolution satellite imagery or radar data, applying various algorithms to first identify surface water and then determine when and where to classify this as flooding. To analyze impacts of floods, units such as communities or households are defined as exposed to floods based on spatial proximity to measured flood incidence.

Most of the economics literature on the household effects of flood exposure uses a single flood incidence measure and does not engage critically with flood remote sensing literature, a criticism that has also been applied to economists’ use of nighttime lights data (Gibson et al., 2025). Guiteras et al. (2015) were among the first (economists) to highlight that different flood measures were poorly correlated, considering rainfall, satellite imagery, and survey reports in Bangladesh, but did not test flood impacts. Chen et al. (2017) find higher correlations between a more refined rainfall-based measure and a measure based on satellite imagery in Bangladesh, and estimate similar qualitative effects of flood exposure on out-migration. Akter (2021) considers survey reports and satellite imagery detection of floods in Pakistan and finds similar qualitative effects on men’s and women’s time use. In contrast, Saunders et al. (2025) report that choosing between measures of exposure based on satellite imagery (including state-of-practice and state-of-the-art models), a physical inundation model, rainfall deviations, and river gauges significantly affects the accuracy and timeliness of simulated flood index insurance payouts in Bangladesh.

In closely related work, Bangalore et al. (2026) compare different approaches to measuring exposure to Nigeria’s 2012 floods. They evaluate three GHSP survey-based measures, two measures based on satellite imagery, one based on media and government reports (the Dartmouth Flood Observatory (DFO) Archive; Brakenridge, 2023), and one combining DFO flood events with satellite

¹⁰Biscaye et al. (2026) provide a summary and overview of current approaches to flood measurement in empirical research. Government and media reports can provide validated information on the timing and location of floods and are incorporated in many public databases of flood events such as the Dartmouth Flood Observatory (DFO, Brakenridge, 2023) and the EM-DAT International Disaster Database (Guha-Sapir et al., 2023). But media sources typically only approximate the spatial extent of flooding and are not comprehensive, with particularly limited coverage outside more populated and developed areas (Jones et al., 2023; Patel, 2024), while the level of detail in government flood records is highly variable. Survey reports provide direct evidence of areas affected by floods but are limited in their spatial and temporal coverage and subject to concerns around measurement error and reference dependence or adaptation. Local precipitation may be correlated with flooding, though the quality of the proxy is questionable (Chen et al., 2017; Guiteras et al., 2015; James and Schumacher, 2024), as most flood events result from the overflow of rivers and other bodies of water (fluvial floods) rather than local rainfall accumulation that is not absorbed or drained away (pluvial floods). Precipitation data may also be combined with hydrological models to predict flood incidence. Water gauges are limited in spatial extent and are more commonly used for validation of other flood measures than for measuring flood incidence. Satellite data sources have the advantages of high spatial and temporal coverage but may fail to capture floods that are of short duration, limited in extent, or masked by clouds (an issue for imagery but not radar), buildings, trees, or topography. They are also sensitive to algorithm parameters selected to identify surface water and classify flood events.

imagery (the Global Flood Database; Tellman et al., 2021). The study finds significant disagreement across sources in which GHSP communities are identified as experiencing floods in 2012, and this leads to inconsistent estimates of the short-term effects of these floods on household agricultural production. We build on Bangalore et al. (2026) and other papers considering approaches to flood measurement by analyzing the drivers of disagreement between survey and satellite measures of community flood exposure in Nigeria in 2012, and explore the implications of flood measurement approaches to estimating long-term effects of the 2012 floods.

First, we use a *survey-based* measure of flood exposure using household and community informant reports in the 2012-13 GHSP.¹¹ We combine reports from separate questions in different parts of the household survey on whether floods in 2012 caused harvest failures, loss of harvest, loss of property, and food insecurity, together with a question from the community survey on events that made people in the community worse off. We define communities with any household or community survey report of flooding in 2012 as exposed to flooding. This approach has the advantage of capturing flood events that affect households in different ways. The household flooding questions focus on only specific dimensions through which floods could affect households, but including the more general community flooding question will help to capture a broader set of flood events including floods that affected households other than those sampled in the GHSP. Importantly, this survey measure will not capture floods which may have occurred near communities without adversely affecting community households: it only identifies reported harmful exposure. However, this is arguably the form of exposure whose consequences are most important to understand. We test the sensitivity of the results to different survey-based flood exposure measures, including measures that exclude different sets of survey flood reports. [Online Appendix C](#) provides additional details on the specific survey questions used and correlations in flood reports across survey questions.

Second, we follow a large number of economics papers (Akter, 2021; Chen and Mueller, 2019; Chen et al., 2017; Guiteras et al., 2015; Sajid and Bevis, 2021; Vitellozzi and Giannelli, 2023) in using a *satellite-based* flood exposure measure using MODIS (Moderate Resolution Imaging Spectroradiometer) satellite imagery.¹² In particular, we use NASA’s 2-day 250 m resolution Global Flood Product (NASA, 2024), taking their approach to identifying flood events as given.¹³ We define pixels as having been exposed to floods if the database identifies any flood incidence in that pixel in 2012 ([Figure A3](#) Panel G presents a map). We match GHSP communities to satellite flooding based on the community coordinates. These coordinates are randomly offset from the original community centroids in the public data, by 0-2 km in urban areas and by 0-5 km in rural

¹¹See e.g., Freudenreich and Kebede (2022), Gray and Mueller (2012), Mueller and Quisumbing (2011), and Stein and Weisser (2022) for examples of research on the household impacts of floods measured using survey reports.

¹²Recent studies using remotely-sensed flood measures increasingly rely on satellites with radar sensors, such as Sentinel-1. The Sentinel-1 radar data are higher resolution than the MODIS data and the radar has the ability to see through clouds and at night, but this satellite was launched in late 2014, after the 2012 flood event we study.

¹³The algorithm identifies surface water using the Normalized Difference Water Index (NDWI), applies a permanent water mask to determine when this water is not expected, and then applies various corrections to deal with cloud and terrain shadow and other known issues in surface water detection. The NASA Global Flood Product transitioned to a different format in 2022 and access to historical flood data has been restricted and is no longer possible through the online platform.

areas (and up to 10 km for a random 1% of rural communities). We therefore define a community as exposed to satellite-detected flooding if its offset coordinates are within 5 km of any flooded pixel, and test robustness to using a 10 km radius. This measure therefore captures proximity to physical inundation. Moving forward we refer to this as the ‘MODIS’ or ‘satellite’ flood measure. his

In addition, we consider two sources of coarse aggregate flood reports. The DFO Archive identifies broad polygons of areas where floods are reported in media.¹⁴ We also directly obtained a list of local government areas (LGAs) – the administrative level below the state – where floods were reported in 2012 from the Nigerian National Emergency Management Agency (NEMA). NEMA is responsible for overseeing disaster management in Nigeria, and collects information on flood reports from State Emergency Management Agencies, other government agencies, and media sources. The 2012 floods were the first event where NEMA attempted to collect information on flooded areas across the country, and resources and tracking efforts were limited relative to subsequent years. We do not directly use this two sources in the analysis of flood impacts because of the coarseness of the flood identification, but to the extent they captures zones where some flooding definitely occurred we analyze how they correlate with survey vs. satellite measures. We do not include a precipitation-based proxy of flood incidence due to the coarseness of publicly-available databases of such proxies, and limitations in directly using higher-resolution local precipitation deviations to identify flooding.

3.3 Empirical approach

We analyze the impacts of community exposure to floods in 2012 on household outcomes in subsequent years in a difference-in-differences framework. Specifically, we estimate two-way fixed effects regressions

$$Y_{icst} = \alpha + \beta Flood_c \times Post_t + \mu_i + \gamma_{st} + \epsilon_{icst} \quad (1)$$

where outcome Y varies across households i , communities c , states s , and years t .¹⁵ $Flood$ is an indicator for having been a resident in 2012 of a community exposed to floods in 2012 according to a particular flood measure. Defining flood exposure at the community level avoids concerns about endogeneity in which households are directly affected. Vulnerability to floods varies across households even within a community: on average, 24% of households report being affected by floods in communities with any survey flood reports (of ten total sample households per community). Analyzing floods at the community level in an ‘intent to treat’ specification captures both direct effects from households that were themselves flooded as well as any indirect impacts from proximity

¹⁴The EM-DAT International Disaster Database, which draws on media and government reports, only records a subset of the states in which floods were reported in 2012.

¹⁵As we consider many outcomes, we calculate Anderson (2008) adjusted q -values to correct for false discovery rates (FDR) with multiple hypothesis tests. These results are available upon request. In general, treatment effects that are significant at more than a 95% confidence level remain statistically significant after this adjustment, while other estimated effects lose statistical significance.

to flooding, though we cannot distinguish between these types of effects.¹⁶

Post is an indicator for being observed in a survey round after 2011. The major flood events in 2012 largely occurred between August and November, at the same time as the 2012 GHSP post-planting survey.¹⁷ As the primary outcomes we consider are reported in the post-harvest survey conducted in February-April 2013, we consider the 2012-13 round (along with the 2015-16 and 2018-19 rounds) to be post-flood. We use data from 2010-11 survey round to serve as a pre-flood comparison period and to test for baseline balance in household and community characteristics. In addition to the average effects over all post-2012 survey rounds estimated in the above specification, we also estimate dynamic effects by interacting *Flood* with indicators for each survey round.

We include household fixed effects (μ) to control for time-invariant household characteristics which might affect livelihood decisions and 2012 flood exposure, and state-by-round fixed effects (γ) to control for common changes over time across broad geographic areas. We test the robustness of the results to varying the set of fixed effects and adding different controls. We cluster standard errors at the level of the community of residence in 2012 since this is the level at which the flood treatment is assigned.

Identification comes from differences in outcomes over time between households in communities that were and were not exposed to flooding in 2012, conditional on fixed differences across households and common shocks at the state-by-survey-round level. We do not assume that flood exposure is randomly assigned. Instead, we interpret estimates as reflecting variation in flood exposure conditional on observable and time-invariant differences across communities and households.

The key assumption of this model is the parallel trends assumption, that trends in outcomes over time would have been the same across communities if it had not been for the floods, conditional on the fixed effects. If the occurrence of floods in 2012 were random over space, this assumption would hold in expectation, but while the timing of floods is exogenous their incidence in space is endogenous. The main threat to identification is that communities at greater risk of flooding have different trends in outcomes even in the absence of flood exposure. The household fixed effects will absorb time invariant characteristics driving flood incidence such as proximity to a river, elevation, slope, soil type, and other hydrological features. We use two approaches to further address potential endogeneity in flood exposure: balancing the sample on recent reported flood exposure and underlying flood hazard, and recentring 2012 flood exposure around predicted 2012 exposure.

To balance the sample, we count the number of years each community was exposed to floods from 2007-2011 using household recall data, and for each strata of years we trim the sample to only include communities within the common support of the distribution of underlying flood hazard

¹⁶Household outcomes may be affected by proximity to flooding even without reporting direct physical effects through local infrastructure damage or expectations of flood risk, for example. Stein and Weisser (2022) find that households living in close proximity to a flood event have lower subjective well-being even if they do not report being directly affected by the flood. We therefore do not use community-level flooding to instrument for household-level flooding due to concerns about violations of the exclusion restriction.

¹⁷One-third of sample households in the 2012-13 post-planting round were surveyed in September and 60% were surveyed in October, with the few remaining households surveyed just before or after.

(from the JRC data) by 2012 flood exposure. We next predict 2012 community flood exposure according to different measures as a function of geographic, weather, and mean household characteristics, using a Lasso regression with 10-fold cross-validation (Table A1 shows the results),¹⁸ and drop communities with predicted exposure values outside the common support of the distribution. This balancing excludes communities for which there is no close 2012 flood counterfactual in terms of flood risk.

We then recenter the 2012 flood treatment variables around estimates of their expectation, following the Borusyak and Hull (2023) approach to dealing with non-random exposure to exogenous shocks. As we do not observe the distribution of potential 2012 flood realizations, we proxy the expected realization with the predicted flood exposure measures. We subtract the predicted flood probability from actual flood exposure to obtain the recentered (or residualized) treatment variable. Impacts are therefore estimated from variation in actual flood exposure among communities with similar predicted levels of exposure based on observed characteristics. As an example, Figure A2 maps communities in the analysis sample colored by the recentered 2012 flood shock variable for the main survey definition. This recentering may be particularly important for survey-reported flood exposure which depends on households being negatively affected by floods. Directly controlling for the predicted probability of reporting such negative effects means we do not just identify effects of floods among particularly vulnerable communities.

In Subsection 5.1, we show that this recentering procedure eliminates baseline differences across exposed and unexposed communities and present placebo and permutation tests that indicate estimated effects are unlikely to be driven by differential trends across areas with different levels of flood risk. We also show that the results are largely robust to different sample restrictions and approaches to controlling for correlates of 2012 flood exposure, suggesting there is overlap in the endogeneity in flood exposure that is captured by the different methods we apply.

4 Results: Flood detection and impacts across sources

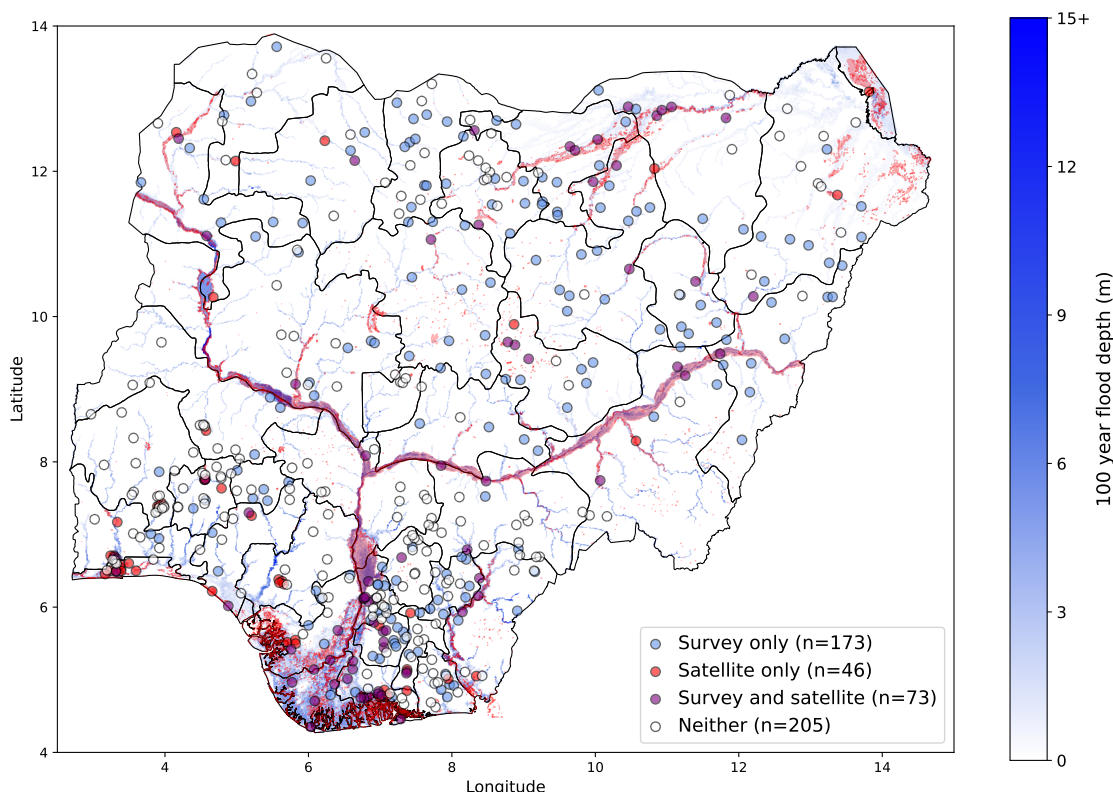
We document three key facts about flood measurement in this setting. First, survey- and satellite-based flood measures disagree substantially and identify flooding in communities with different characteristics. Second, the disagreement can be attributed to differences in both the nature of flooding that is captured and in measurement limitations. Third, the survey measure reveals severe negative livelihood impacts of 2012 community flood exposure while the satellite measure shows attenuated or even positive estimated effects.

¹⁸The predicted 2012 flood likelihood does not come from a full hydrological model and may depend on the selection of predictors. The most common important predictors are distance to the nearest body of water, average 100-year flood depth, slope, measures of land cover and particularly agricultural land, and household reports of floods over the previous five years. Interactions between 2012 rainfall deviations and both elevation and agricultural area, as well as average wettest quarter rainfall, are also selected in most predictions. Other rainfall predictors including interactions of 2012 rainfall deviations with several other geophysical characteristics are not commonly selected, consistent with prior evidence on the weak relationship between local rainfall and flood incidence. Including 2012 rainfall in the predictions further helps to isolate impacts of inundation from those of heavy rainfall.

4.1 Differences in flood detection

The set of communities identified as exposed to floods in 2012 differs significantly depending on the measure used. [Figure 1](#) maps the locations of GHSP communities and their 2012 flood exposure status based on the main survey measure and the satellite measure. The figure highlights substantial differences in community flood exposure between the two measures. Of 497 GHSP communities, 246 are identified as flooded by the survey measure and 119 by the satellite measure, but only 73 are identified by both.

Figure 1: 2012 community flood exposure, by flood measure

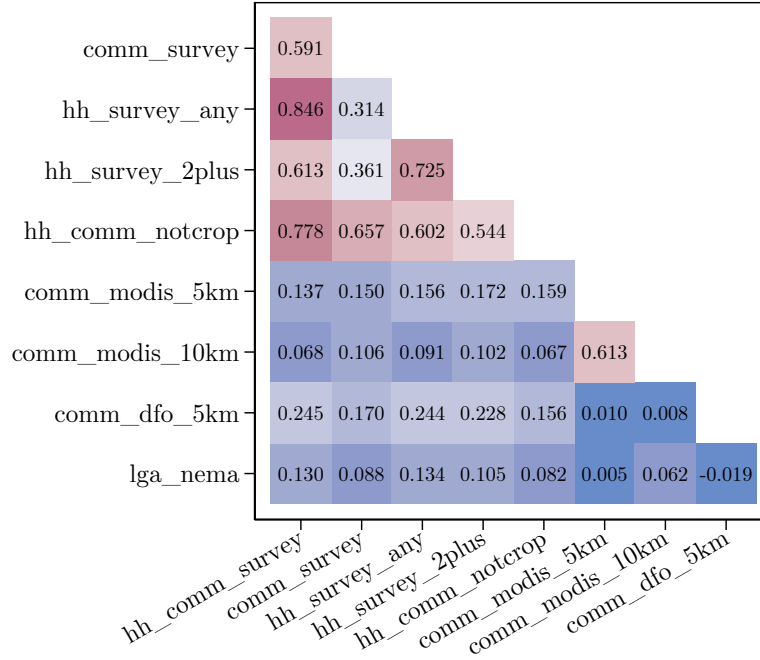


Note: The map shows the locations of communities in the Nigeria GHSP sample, colored by their flood exposure in 2012. Flood exposure is based on either any household or community survey report of being affected by floods in the community or on the community being within 5 km of a pixel where flooding was identified in the NASA MODIS-based flood tracking product. Pixels identified as flooded by the NASA product are shown in red. The blue shading in the background corresponds to the depth of 100-year fluvial floods, which largely tracks major rivers and bodies of water.

[Figure 2](#) shows a heatmap of pairwise correlations between flood measures ([Figure A3](#) maps exposed communities by definition). Survey flood reports come from multiple types of questions, but survey-based flood measures focusing on different subsets of questions are moderately to highly correlated with each other.¹⁹ In contrast, correlations between survey measures and the satellite-based measure are weak (all below 0.17), and even more so with a 10 km buffer suggesting that larger buffers introduce additional misclassification. This indicates that the two approaches identify largely different sets of communities as exposed to flooding.

¹⁹Household floods reports are more common (205 communities with at least one) than community informant reports (127) and 44% of households reporting a flood are in communities with no community informant flood report,

Figure 2: Correlations between different measures of community-level 2012 flood exposure



Note: This heatmap shows pairwise correlations between different measures of flood exposure in 2012 at the level of communities in the Nigeria GHSP. ‘hh_comm_survey’ is the main survey-based measure and indicates any flood report by either a household or community informants. ‘comm_survey’ indicates a flood report by informants in the community survey. ‘hh_survey_any’ indicates any household flood report in any survey module. ‘hh_survey_2plus’ indicates at least two households reporting a flood. ‘hh_comm_notcrop’ indicates any community informant flood report or any household report of a flood excluding reports of flood-related crop destruction or damage. ‘comm_modis_5km’ and ‘comm_modis_10km’ indicate any flooded pixel identified in the NASA MODIS flood database within 5 and 10 km of the community centroid, respectively. ‘comm_dfo_5km’ indicates whether the community is within 5 km of a polygon of flooding recorded in the Dartmouth Flood Observatory archive. ‘lga_nema’ indicates floods reported by NEMA in the local government area (LGA) containing the community.

Potential validation data are limited, but we compare the correlations between survey and satellite floods measures and DFO and NEMA flood records. Both sources identify a selected set of flooded areas. The DFO Archive identifies five broad polygons affected by floods covering 56 GHSP communities, while the NEMA records list LGAs covering 99 communities. While these validation sources are coarse and do not capture local variation, their stronger correlation with survey-based measures – and essentially zero correlation with the satellite measure – suggests that survey reports are more closely aligned with damaging household flood exposure in this setting.

To test whether there are systematic differences in the types of communities identified by the main survey and satellite flood measures, we estimate a series of regressions of community characteristics on dummies for flood exposure under different combinations of the two measures, and present the results in [Table 1](#). Communities identified as flooded by both measures are located close to water bodies, face high levels of fluvial flood hazard (depths of 100-year floods), and experienced unusually low precipitation in 2012 relative to historical averages. As might be expected given fluvial floods are the main form for flood risk in most of Nigeria, areas that are not classified as flooded according to either flood measure are higher in elevation and farther from a body of water,

suggesting those informants lack full information about flooding within the community.

face very little fluvial flood hazard, and have a greater slope (column β_3).

Table 1: Differences in GHSP community characteristics by 2012 flooding status

	Mean: Both definitions N=73	β_1 : Satellite only N=46	β_2 : Survey only N=173	β_3 : No flood N=205	$\beta_1 = \beta_2$ (<i>p-val</i>)	$\beta_1 = \beta_3$ (<i>p-val</i>)	$\beta_2 = \beta_3$ (<i>p-val</i>)
Rural	0.63 (0.49)	-0.37*** (0.08)	0.17*** (0.06)	0.06 (0.06)	0.00***	0.00***	0.01**
HH distance to nearest major road (km)	7.23 (11.21)	-3.43** (1.54)	0.03 (1.14)	-2.01* (1.11)	0.01***	0.22	0.01**
Elevation (m)	197.74 (269.85)	-38.41 (38.49)	134.76*** (28.54)	73.23*** (27.87)	0.00***	0.00***	0.00***
Slope (percent)	2.32 (1.44)	0.35 (0.44)	0.63* (0.32)	1.28*** (0.31)	0.44	0.02**	0.01**
Distance to nearest water area (km)	12.86 (18.99)	9.61* (5.31)	18.00*** (3.94)	30.46*** (3.85)	0.06*	0.00***	0.00***
Avg 100-year flood depth within 5 km	1.28 (1.41)	-0.67*** (0.13)	-1.12*** (0.09)	-1.16*** (0.09)	0.00***	0.00***	0.37
Diff. in annual precipitation from NOAA CPC hist. avg. (mm)	-306.96 (586.52)	18.52 (72.26)	313.33*** (53.57)	213.12*** (52.32)	0.00***	0.00***	0.00***
Baseline share HHs w/ >5 members	0.46 (0.25)	-0.10** (0.04)	0.07** (0.03)	-0.03 (0.03)	0.00***	0.04**	0.00***
Baseline share HHs below 1.90 USD/cap poverty line	0.31 (0.31)	-0.14*** (0.05)	0.15*** (0.04)	0.02 (0.04)	0.00***	0.00***	0.00***
Baseline share HHs w/ above-median asset value	0.56 (0.27)	-0.02 (0.05)	-0.09** (0.04)	-0.07* (0.04)	0.13	0.32	0.42
Baseline mean HH agricultural land (ha)	0.73 (1.51)	-0.50 (0.37)	0.31 (0.27)	0.06 (0.27)	0.01**	0.08*	0.24
Baseline share HHs w/ crop production	0.60 (0.40)	-0.29*** (0.06)	0.26*** (0.05)	0.08* (0.05)	0.00***	0.00***	0.00***
Baseline share HHs w/ non-farm enterprise	0.78 (0.17)	-0.01 (0.04)	-0.03 (0.03)	-0.01 (0.03)	0.56	0.83	0.55
Baseline share HHs w/ wage employment	0.45 (0.27)	-0.02 (0.05)	-0.08** (0.03)	-0.04 (0.03)	0.13	0.56	0.15

Note: This table presents the results from regressions of community characteristics (in each row) on dummies for flood exposure in 2012 according to different measures. The reference group is communities considered flooded under both the survey-based and satellite-based measures. Baseline household data are from the GHSP 2010-11 survey round. The difference in annual precipitation from the historical average is the total precipitation for 2012 minus the annual average from 1960-1990.

Communities identified as flooded by the satellite measure only (column β_1) are more urban, less agricultural, and face lower underlying flood hazard. Households in these areas are less likely to be poor or engaged in crop production. These differences suggest that satellite-based measures are more likely to capture flooding in settings where economic damages may be more limited or more easily mitigated. In communities where both measures or only satellites identify flooding, the significantly lower precipitation relative to historical averages (which represents a 19% difference) also suggests a potential role of cloud cover in determining where satellites can detect floods. In contrast, communities identified as flooded only in the survey (column β_2) experienced no decrease in 2012 precipitation relative to historical averages. These communities are also located farther from major water bodies and are more rural and more agricultural, and households are more vulnerable to damages from floods. Further analysis of predictors of flood exposure confirms that the two measures capture different types of events (Table A1). While both are associated with underlying flood hazard, the satellite measure is more strongly tied to fluvial flood risk whereas the survey measure is strongly predicted by rainfall intensity in 2012.

4.2 Explaining survey-satellite disagreement

A key limitation in comparing survey and satellite flood measures is that we do not observe exact community locations. Random displacement of GHSP coordinates introduces error in matching communities to nearby satellite-detected flooding. As a result, some apparent disagreements may reflect misalignment between reported and true locations. To assess the likely magnitude of this issue, we simulate LSMS-style coordinate displacement and then rematch communities to satellite-detected floods using the same 5 km buffer approach. These simulations suggest that coordinate displacement has only a modest effect on survey-satellite overlap (Figure A5). The original public coordinates classify 173 communities as survey-only flooded and 73 as flooded by both survey and satellite measures. Across 1,000 simulated displacements, the median counts are 171 and 75, respectively, with 95% of simulations classifying between 164 and 177 communities as survey-only flooded and between 69 and 82 as flooded by both measures. Therefore, while 47 of the 173 survey-only communities are within 10 km of a satellite-detected flood and could potentially have been within 5 km in reality (Figure A4), the displacement simulations show that plausible coordinate jittering does not substantially reduce the number of survey-only communities, indicating that other factors play a larger role.

Differences in how flooding is defined also contribute to disagreement. Survey-based measures identify floods that adversely affect households, particularly through crop loss, property damage, or food insecurity. In contrast, satellite measures capture changes in surface water regardless of economic impact. Table 1 shows that communities identified as flooded only by the satellite measure are less rural, less agricultural, and less vulnerable, suggesting that households are less likely to experience and report adverse effects of detected surface water changes. Conversely, communities identified only by the survey measure are more rural, poorer, more agricultural, and more exposed to heavy rainfall, consistent with survey reports capturing pluvial flooding that is difficult for MODIS imagery to detect.

In addition to differences in definitions of flooding, the two measures are also subject to different types of measurement error which could create disagreement. Survey measurement error appears unlikely to explain much of the disagreement. Households may report flooding that affected them but which took place away from the community, or may strategically misreport flood exposure. But requiring at least two households to report flooding yields similar correlations with the satellite measure, and the correlation is also similar when exclusively considering community survey flood reports. The share of households reporting flooding is also similar across communities identified by survey only and by both survey and satellite.

On the other hand, limitations of the satellite measure are likely to be very important in this setting.²⁰ First, the satellites only take one image every 1-2 days so will miss short-lived or flash flooding events. Second, missing data due to cloud cover is substantial. The median (mean) MODIS pixel in Nigeria was cloudy for 71% (73%) of images from August-November (the main flooding period in 2012), and the median across GHSP communities for the daily share of pixels within 5 km

²⁰Online Appendix D presents more detail and examples of the limitations of MODIS imagery for flood detection.

masked by clouds was 100% for nearly all of August-October. Third, the 250 m spatial resolution limits detection of localized flooding, as detection requires a large share of a pixel to be covered by water.²¹ Fourth, the flood detection algorithm may misclassify some expected or seasonal variation in surface water as flooding.²² The result of these limitations is that the satellite measure misses a large number of economically relevant flood events.²³

In summary, disagreement between the two flood measures reflects both differences in how flooding is defined and limitations in measurement. Satellite-based measures provide an objective indication of surface water but may misclassify seasonal variation as flooding and are heavily constrained by cloud cover and spatial resolution. Survey-based measures capture economically meaningful flood exposure but are subjective and may depend on vulnerability to floods. We estimate household impacts of flood exposure from both measures, but given the limitations we identify in the satellite-based measure we focus on survey-based flooding in subsequent analysis.

4.3 Implications of flood measurement decisions for estimated impacts

Table 2 and Table 3 present the estimated impacts of 2012 flood exposure using the survey- and satellite-based flood measures, respectively. We estimate both average effects of flood exposure over the following three survey rounds and dynamic effects in each round. In each table, the rows indicate the outcome variable and the columns indicate the timing of the estimated flood exposure effect. Both flood measures are recentered around their predicted values, so coefficients represent the estimated effect on households of having resided in 2012 in a community that was exposed to flooding according to a particular measure relative to households in non-flooded communities with the same predicted 2012 flood propensity in the same state.

Under the main survey-based flooding measure (Table 2), we find that – for a given predicted level of exposure – any reported 2012 flood exposure in a community decreases household daily consumption per capita by 0.65 USD (12%) and total farm production value by 313 USD (17%) on average over the following three survey rounds, and increases the count of household members in wage employment by 0.14 (17%). The decreases in consumption per capita are largest in 2012-13 immediately after the floods and lose significance by the 2018-19 round. This effect is driven entirely by food consumption with no significant effects on non-food consumption, is similar but more stable over time if we consider consumption per adult equivalent, and increases the share of households below the international poverty line by 16% on average (Table A12). Despite the decrease in consumption and particularly food consumption, we find no significant average effect on the likelihood the household reports experiencing food insecurity, and in 2018-19 there is a significant decrease in this likelihood.

²¹Higher-resolution satellite radar would help overcome limitations based on cloud cover and resolution, but such data are not available for 2012 and the tradeoff is a lower revisit frequency.

²²Slayback (2024) notes that the NASA Global Flood Product represents a “best guess” as to the flood status of detected water.

²³These limitations also apply to the remote detection of other spatial features. For example, Aker et al. (2026) find high levels of error in detection of agricultural demi-lunes in Niger using 30cm and 10m satellite imagery and point to challenges in image quality, availability, cloud cover, and resolution.

Table 2: Impacts of survey-reported 2012 community flood exposure

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Any food insecurity in last 12 months	11691	0.17 [0.38]	-0.02 (0.02)	0.00 (0.03)	-0.03 (0.02)	-0.10** (0.05)
Daily HH consumption per capita (USD)	11691	5.64 [4.40]	-0.65*** (0.22)	-0.83*** (0.30)	-0.47* (0.25)	-0.46 (0.30)
Total value of HH farm production (USD 100s)	11691	18.12 [40.83]	-3.13* (1.74)	-4.02** (1.88)	-2.17 (2.00)	-2.92 (2.72)
Total non-farm HH enterprise earnings (USD 100s)	11691	26.83 [77.46]	-4.00 (3.01)	-1.81 (3.26)	-6.97** (3.44)	-1.32 (5.17)
Total wage employment earnings (USD 100s)	11691	40.63 [188.67]	4.69 (5.39)	7.48 (6.40)	2.25 (5.45)	1.02 (10.08)
HH farm share of total HH earnings and prod value	11328	0.43 [0.44]	-0.00 (0.01)	-0.00 (0.02)	-0.01 (0.02)	0.01 (0.03)
NFE share of total HH earnings and prod value	11328	0.35 [0.41]	0.01 (0.02)	0.01 (0.02)	0.02 (0.02)	-0.01 (0.03)
Wage share of total HH earnings and prod value	11328	0.18 [0.35]	-0.02 (0.01)	-0.02 (0.01)	-0.02* (0.01)	0.00 (0.02)
HH earnings Herfindahl-Hirschman Index	11328	0.85 [0.20]	0.00 (0.01)	0.00 (0.01)	0.01 (0.01)	-0.01 (0.02)
Count of HH members wkg. in HH farm	11691	1.78 [2.09]	0.08 (0.09)	0.08 (0.09)	0.07 (0.11)	0.16 (0.19)
Count of HH members wkg. in HH non-farm enterprise	11691	1.21 [1.26]	0.06 (0.06)	0.02 (0.05)	0.08 (0.07)	0.15 (0.12)
Count of HH members wkg. in wage employment	11691	0.82 [1.40]	0.14*** (0.05)	0.14** (0.06)	0.13** (0.06)	0.25*** (0.08)
Any member left HH for work during/since last round	11691	0.02 [0.13]	-0.01 (0.01)	-0.01 (0.01)	-0.03** (0.02)	0.06* (0.03)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Online Appendix C](#) for details on variable construction. Sample sizes are slightly smaller for the earnings share variables due to some households reporting zero earnings or production value across all activities. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. 2012 flood exposure is recentered around predicted 2012 exposure. Estimates therefore represent the effect of having resided in a community that was exposed to floods 2012 according to survey reports for a given predicted propensity of survey-based exposure. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Point estimates for the effect of flood exposure on the value of farm production are consistently negative and large but are only statistically significant in 2012-13, where we find a 22% decrease. The average effect is driven by a significant 450 USD (30%) decrease in crop production value, offset by a marginally significant 137 USD increase in the value of livestock production ([Table A13](#)). Estimated effects on earnings from household NFE are also negative but noisy, though we find a significant 697 USD (26%) in 2015-16. Point estimates for wage employment earnings are positive but in no case statistically significant. Despite reductions in farm production value, we find no change in the share of total household earnings and production value from different activities besides a decrease in the share of wage earnings in 2015-16, and no change in the Herfindahl-Hirschman Index (HHI) of concentration of earnings across activities. On average, household earnings shares appear quite diversified, with 43% to farm production, 35% to NFEs, 18% to wage employment, and 4% to other sources. But the mean household HHI value of 0.85 – where 1 indicates all earnings

or production value are from a single activity – reveals much greater concentration at the household level. The mean of the maximum earnings share across all activities is 88% for sample households in 2010-11, and the median is 98%. While earnings shares are diversified across households on average, most individual households are highly specialized even if engaging in multiple livelihood activities is common, and this does not change after flood exposure.

We find positive point estimates for the effect of the 2012 floods on the count of household members engaged in different livelihood activities, but only for wage employment are these effects statistically significant. Households in flood-exposed communities increase the count of members working for a wage in every round after 2012, with the largest increase in 2018-19. The likelihood that any household member migrates for work does not change on average, but this masks differences in effects over time. Flood-exposed households are 3 percentage points (pp) less likely to have any member leave between survey rounds in 2015-16, but 6 pp *more* likely in 2018-19.

We compare the average impacts of flood exposure across different survey-based measures and find broadly similar results (Table A2). Effects on consumption per capita are large and statistically significant across survey definitions, but small differences in magnitudes and precision highlight the importance of considering specific ways that surveys ask about flood exposure. Effects on the count of household wage workers remain large but lose statistical significance if we omit community survey reports from the flood definition, suggesting these community reports help to capture communities that suffered flood damages that fell short of the flood-related shocks asked about in the household surveys but that nevertheless adversely affected households. We find larger effects on value of farm production if we require two household flood reports, consistent with more severe impacts for households reporting flood-related damages. We measure flood exposure in part based on household reports on crop failure or losses. This should mechanically lead to impacts on crop production, though this relationship is not direct as we define flood exposure at the community level which includes many households not reporting any direct effects of floods and also consider flood reports unrelated to crop damage. Further, communities where households experience crop damages but not failures and no other flood impacts are reported would be considered non-flooded due to how the question on crop failures due to flooding is worded, attenuating estimated impacts on crop production. If we eliminate communities where the only flood reports are of crop failure from the sample we find a similar magnitude but noisier effect on farm production value and other impacts remain significant. This indicates that floods adversely affect households even where they do not result in crop failure.

We find a very different pattern of results if we use a satellite-based flood measure (Table 3).²⁴ We find a larger average decrease in farm production value (378 USD, 20%), but this is surprisingly driven by decreases in 2015-16 and 2018-19. Estimated effects in 2012-13, when farm production

²⁴Note that the sample sizes using the satellite-based measure are smaller because we restrict the sample to communities within the range of common support of 2012 predicted flood exposure by actual satellite-measured exposure. This restriction eliminates more communities when using the satellite-based measure than when using the survey-based measure. Sample composition differences do not explain the divergence between survey- and satellite-based estimates. Re-estimating both specifications on a common sample of communities with non-missing values for both recentered flood measures yields qualitatively similar results (Table A3).

would have been directly affected by floods, are smaller and not statistically significant. In contrast to the survey measure, satellite-based floods *increase* earnings from NFEs by 1,214 USD (45%) on average, with large increases in both 2012-13 and 2015-16. This is associated with a significant increase in the share of total earnings from NFE in 2012-13 only. The earnings HHI is also significantly larger in 2012-13, indicating more concentrated incomes in the year of the floods. Wage earnings and the count of household members in different activities do not change, except for an increase in the count of household members engaged in farm work in the 2018-19 round and a decrease in the likelihood of any member having left the household in 2018-19. Importantly, we find no average effect of the floods on consumption per capita. The point estimates are positive on average and even in 2012-13, negative but small in 2015-16, and large, positive, and significant in 2018-19.

Table 3: Impacts of satellite-detected 2012 community flood exposure

	N	Control Mean (SD)	2012 MODIS recentered × Post (SE)	2012 MODIS recentered × 2012-13 (SE)	2012 MODIS recentered × 2015-16 (SE)	2012 MODIS recentered × 2018-19 (SE)
Any food insecurity in last 12 months	8872	0.18 [0.39]	0.02 (0.03)	0.03 (0.03)	0.00 (0.04)	0.08 (0.07)
Daily HH consumption per capita (USD)	8872	5.17 [4.18]	0.20 (0.33)	0.43 (0.46)	-0.15 (0.35)	0.72* (0.43)
Total value of HH farm production (USD 100s)	8872	18.49 [37.36]	-3.78** (1.92)	-2.45 (1.96)	-4.87** (2.38)	-5.61** (2.63)
Total non-farm HH enterprise earnings (USD 100s)	8872	26.69 [79.41]	12.14*** (4.63)	12.68** (5.13)	13.60*** (5.02)	1.85 (6.49)
Total wage employment earnings (USD 100s)	8872	34.38 [162.08]	-1.05 (8.86)	2.44 (10.24)	-5.94 (9.11)	4.50 (17.77)
HH farm share of total HH earnings and prod value	8569	0.47 [0.43]	-0.02 (0.02)	-0.03 (0.02)	-0.02 (0.02)	-0.01 (0.03)
NFE share of total HH earnings and prod value	8569	0.33 [0.39]	0.03 (0.02)	0.05* (0.02)	0.02 (0.02)	0.06 (0.04)
Wage share of total HH earnings and prod value	8569	0.17 [0.34]	-0.01 (0.02)	-0.00 (0.02)	-0.02 (0.02)	-0.03 (0.03)
HH earnings Herfindahl-Hirschman Index	8569	0.84 [0.20]	0.01 (0.01)	0.03** (0.01)	0.00 (0.01)	-0.02 (0.03)
Count of HH members wkg. in HH farm	8872	2.01 [2.21]	-0.02 (0.11)	-0.07 (0.11)	-0.07 (0.15)	0.49*** (0.18)
Count of HH members wkg. in HH non-farm enterprise	8872	1.28 [1.33]	0.04 (0.05)	0.04 (0.06)	0.00 (0.07)	0.22 (0.14)
Count of HH members wkg. in wage employment	8872	0.76 [1.33]	0.03 (0.07)	0.06 (0.09)	-0.02 (0.08)	0.11 (0.08)
Any member left HH for work during/since last round	8872	0.01 [0.11]	-0.02 (0.01)	0.00 (0.01)	-0.03 (0.02)	-0.11*** (0.04)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether the community coordinates are within 5 km of any pixel identified as flooded using MODIS satellite imagery in the NASA Global Flood Product.. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Online Appendix C](#) for details on variable construction. Sample sizes are slightly smaller for the earnings share variables due to some households reporting zero earnings or production value across all activities. Control means are for communities with no 2012 satellite-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. 2012 flood exposure is recentered around predicted 2012 exposure. Estimates therefore represent the effect of having resided in a community that was exposed to floods 2012 according to satellite detection for a given predicted propensity of satellite-based exposure. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The estimated effects using a satellite-based measure are highly sensitive to the choice of buffer.

Using a 10 km buffer to define community exposure to satellite-detected flooding, we find no effects on consumption, farm production value, or NFE earnings, or any other significant effects besides a decrease in the count of household wage workers (Table A2). It is difficult to interpret what effects this 10 km buffer definition is identifying given its very low correlation with all survey and other measures of 2012 flood exposure (Figure 2). The 10 km buffer measure includes 47 more communities with survey-reported floods compared to the 5 km measure, but also 60 more communities with no survey-reported floods.

The difference in results between the main survey- and satellite-based definitions, if taken at face value, would lead to very different conclusions about the effects of flood exposure. Survey-based measures generally agree that 2012 flood exposure makes households worse off in terms of consumption per capita on average over several following years, and that this is associated with large decreases in the value of farm production that are not offset by earnings from increased labor supply in wage employment. In contrast, the 5 km satellite buffer measure would suggest that households are largely resilient to flood shocks and offset decreases in farm production value by greatly increasing household NFE earnings, though without meaningfully changing labor supply to different activities. Strikingly, the satellite measure indicates that exposure to the 2012 floods made households better off in the very short-term, by increasing NFE earnings without significantly harming farm production or consumption.

Significant impacts on consumption and farm production under the survey measure are consistent with this measure directly identifying economically-relevant flood exposure: it is conditional on some type of realized adverse impact. But it was not *a priori* obvious that defining community-level flood exposure based on such reports would result in significant estimated effects given that only a quarter of households in flooded communities report direct effects. No significant adverse effects in 2012-13 according to the satellite measure reflect that it is more likely to detect floods in more urban and less agricultural areas – potentially less vulnerable to flood damages – and misses 173 communities with survey reports of negative impacts of the floods. Although the survey measure will miss some floods that do not harm households, the fact that the satellite measure of physical inundation fails to capture such a large share of reported harm suggests it is a less reliable measure of exposure to a flood shock in this setting. We next turn to a deeper exploration and analysis of the impacts of flood exposure under the main survey definition.

5 Discussion: Impacts of flood exposure

Whether households are negatively affected by floods when they occur can depend on their characteristics. We first present identification and robustness tests to help evaluate whether the impacts of survey-based flooding that we estimate can be interpreted as causal effects of flood exposure. We then explore the mechanisms behind the effects on farm production value and consumption that we observe. Finally, we test whether impacts of flood exposure depend on baseline community or household characteristics.

5.1 Identifying causal impacts of flood exposure

Our identification strategy estimates effects of the 2012 floods from differences in flooding across communities in the same state with the same predicted exposure and recent survey-reported flooding histories, in a difference-in-differences framework where we observe the same households before and after the 2012 floods. The key identifying assumption is that households in flooded and non-flooded communities would have experienced parallel trends in outcomes in the absence of flooding, conditional on controlling for predicted flood exposure, fixed household and community characteristics, and time-varying state-level shocks.

The parallel trends assumption cannot be tested, and with only one round of survey data before the 2012 floods we also cannot test parallel trends in outcomes pre-exposure. As a partial assessment of the plausibility of the assumption, we test for baseline (2010-11) balance in community and household characteristics by treatment status (Table A4). In line with the fact that flood exposure is not exogenous across space and with Table 1, we find that characteristics of communities and households flooded in 2012 differ from those not flooded along many margins. Most of these differences disappear after recentering the treatment around predicted 2012 flood exposure, and we can no longer reject the joint hypothesis that all characteristics are the same ($p = 0.438$). We test the robustness of the results to including baseline characteristic-by-round controls for the characteristics that remain unbalanced. In addition to balance in baseline household and community characteristics, recentering the treatment variable also leads to balance in annual survey-reported flood exposure across the full survey period (Table A5), reducing concerns about endogeneity.

One potential threat to identification is differential attrition from the panel sample by flood exposure. The final analysis sample using the main survey flood definition after balancing the sample of communities and recentering the flood exposure variable includes 4,018 unique households across 428 communities. We only include households that did not leave their original survey community prior to the 2012-13 survey round. Although movers were tracked in the GHSP surveys, we cannot construct the same community-level flood measures for households outside the main sample communities in 2012. This drops 110 households (2.7%) from the sample, and attrition is actually higher in non-flooded communities (3.5%) compared to flooded communities (1.7%). 5.3% of analysis sample households – observed in both 2010-11 and 2012-13 – dropped out of the survey in 2015-16, but there is no difference by flood exposure ($p = 0.453$). Of the analysis sample households, only 1,188 are observed in the 2018-19 round because of the partial GHSP sample refresh. The lower sample size for 2018-19 leads to larger standard errors for estimated dynamic effects in this period. Inclusion in the panel subsample was randomized at the community level within rural and urban parts of six geopolitical zones, and 134 of the analysis sample communities were retained. All original sample households were targeted in each panel community, and we find that on average 8.9 of the original sample of 10 were successfully surveyed, with slightly higher coverage in communities exposed to the 2012 floods. Together, these results indicate that attrition from the panel was minimal and that exposure to the 2012 floods did not cause differential attrition.

We now turn to tests of the identification strategy. Recentering flood exposure around its pre-

dicted value directly controls for potential differences in trends in communities at different risk from floods. Among non-flooded communities, predicted 2012 flood propensity is associated with differences over time in wage earnings, the count of wage workers, and migration of household members (Table A6), underscoring the importance of isolating the unexpected component of flood exposure. The results are somewhat sensitive to estimating impacts of binary rather than recentered flood exposure (Table A7). A binary exposure model with no added controls and without community sample restrictions yields very similar results as the main specification. This suggests that the household fixed effects account for a large share of time-invariant community characteristics potentially creating different trends in outcomes over time. After adding the main specification sample restrictions, the decrease in consumption per capita is larger in magnitude while the decrease in farm production value is slightly smaller and the effect on the count of household wage workers is smaller and loses significance. Adding average flood hazard-by-round and 2012 rainfall deviations-by-round controls does not meaningfully change the results. Though the results are similar in general, the differences indicate that the recentering approach in the main specification may account for some endogeneity in flood exposure not controlled for by the household fixed effects.

We next conduct a treatment permutation test to evaluate whether the estimated effects identify impacts outside of differences over time by risk of 2012 flood exposure. The permutation test reassigns 2012 flood exposure within bins of predicted 2012 flood propensity while preserving the number of treated communities in each bin. Focusing on daily per capita consumption and crop production value, we find that the resulting distributions of placebo coefficients are centered around zero (Figure A6). The actual estimated effects lie in the extreme tails of these distributions and would be very unlikely under random reassignment of treatment within flood risk bins. We estimate randomization p -values of 0.001 for daily per capita consumption and 0.008 for the value of crop production. These results suggest that the estimated effects are unlikely to be driven primarily by differences in trends across areas more vulnerable to the 2012 floods, consistent with a framework in which identification comes from realized exposure net of predicted exposure risk and changes over time controlling for fixed household and community characteristics.

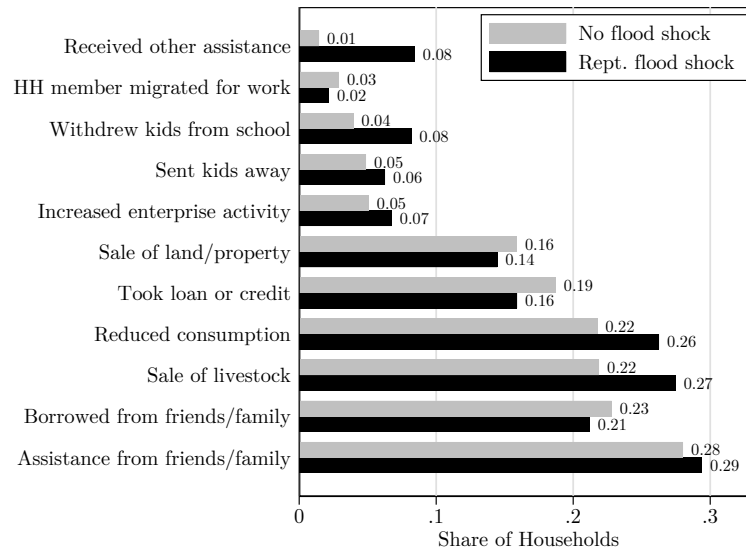
The main results are robust to adding controls and using round fixed effects (Table A8). The magnitudes of the significant effects of flood exposure are larger when including baseline characteristic-by-round controls for characteristics that were unbalanced by recentered flood exposure, and the effect on consumption is smaller while the effect on farm production is larger when shifting from state-by-round fixed effects to round fixed effects. The results are also qualitatively similar with different sample restrictions (Table A9). Results are nearly unchanged if we include all communities or if we only drop communities more than 50 km from any community with a 2012 flood report. This is likely because the construction of the recentered flood shock variable already drops communities outside of the common support of estimated 2012 flood propensity by actual flood exposure, making the additional sample restrictions less meaningful. Decreases in farm production value are larger and more strongly significant if we exclude control communities within 20 km of or within the same LGA as any flooded community, which could potentially have been

indirectly affected by nearby flooding, but other results are similar. These results suggest some localized spillovers of flood exposure but that the main effects are driven by flood exposure within communities. Finally, we show that statistical significance is largely not sensitive to the choice of standard error clustering ([Table A10](#)).

5.2 Decreases in consumption per capita

Exposure to the 2012 floods leads to a large decrease in household per capita consumption, particularly in the year of the floods and driven by reduced food consumption. The reduction in consumption aligns with the stated coping strategies used by households reporting being affected by a flood shock. Just 2.9% of sample households have any formal insurance of any kind in 2012, meaning they must use a variety of strategies to cope with shocks, and reduced consumption is among the most common along with selling livestock and borrowing or receiving assistance from friends and family. [Figure 3](#) compares the strategies reported by households that declare a flood shock in 2012 against those reported by households declaring only non-flood shocks.²⁵

Figure 3: Household shock coping strategies, by reported flood exposure



Note: The figure shows the share of households reporting each type of coping strategy in 2012-2013, separately for households declaring experiencing a flood shock and for households declaring only non-flood shocks. Coping strategies are aggregated across all shocks faced by the household as many shocks are interrelated. [Table A11](#) shows differences in the probability of reporting a given coping strategy by reports of a flood shock controlling for other shocks reported.

²⁵Information on these strategies comes from the household survey shocks module in the 2013 post-harvest survey, where households are asked to report what adverse shocks they have faced and to list the coping strategies they have used. Death or disability of an adult household member is the most commonly declared shock for 2012, followed by floods ([Figure A7](#)). For both groups in [Figure 3](#), we consider all the coping strategies households report across all shocks they face, as many shocks are interrelated. For example, households declaring a flood shock are more likely to also declare non-flood harvest destruction, death of livestock to illness, and adverse price shocks than households declaring shocks other than floods, even after including state fixed effects. We set aside responses of ‘did nothing’ which are the most common response for both groups, at 40% for households reporting a flood shock and 36% for other households.

The coping strategies reported by flooded and non-flooded households appear broadly similar in terms of relative frequency, but with some differences. Controlling for other shocks reported and community fixed effects, households declaring flood shocks are significantly more likely to report sales of livestock and land/property, increasing enterprise activity, reducing consumption, and receiving assistance from outside of family and friends. These suggest that flood shocks may be both more severe and attract more support than other types of household shocks, though even among flooded households just 8% report receiving assistance from outside their immediate network, such as government and non-profit relief. Borrowing and selling assets – among the most common coping strategies – may reduce household wealth in the long term and could drive persistent effects of flood exposure. We do not observe borrowing or land transactions, but find no significant effect of flood exposure on the aggregate reported value of household assets and a marginally significant positive effect on livestock holdings (Table A12), so reported coping through asset sales is not captured by the outcomes we measure.

On the other hand, the 15% decrease in consumption in 2012-13 is consistent with floods representing a severe immediate shock and households using coping strategies that deplete available resources. Although the consumption effect loses significance in 2018-19 ($p = 0.131$), the point estimate is similar to the significant effect in 2015-16. These results indicate that community flood exposure makes households significantly and persistently worse off on average, even accounting for the fact that only 24% of households sampled in flooded communities report any direct adverse impacts of floods. These households may be indirectly affected by local flooding, or affected in ways not captured by the survey questions around flood damages. Alternatively, the average effects we estimate may be driven by much larger effects among the worst-affected households. The decrease in food consumption is not associated with an increase in food insecurity, which may suggest that households substitute more expensive foods for cheaper staples. These results build on previous work on household flood impacts in African countries showing short-term decreases in consumption and food security (Amolegbe et al., 2023; Baez et al., 2020; Devereux, 2007; Reed et al., 2022), and similar results for an analysis of effects of natural disasters in Nigeria (Escalante et al., 2025).

5.3 Changes in livelihood activities

The central question of this paper is whether a weather-related natural disaster – in this case, floods – can promote sustained shifts of livelihood activities away from more vulnerable household agriculture. The literature on floods fairly consistently finds that flood exposure increases short-term engagement in non-farm labor, with some exceptions, but the results in Table 2 suggest limited livelihood effects of flood exposure in this setting. We find no changes in the share or concentration of total earnings and production value from different activities, and increases in the count of wage workers but not in wage earnings.

Table 4 shows the impacts of flood exposure on extensive and intensive margin engagement in different livelihood activities. The increase in the number of household members active in wage employment is driven by changes on the intensive margin. We find no effect on the probability of

any household wage worker but persistent increases in total household hours in wage employment: a significant 15% increase on average for hours in the last 7 days at the time of the post-planting (PP) survey and a noisy 6% increase for the post-harvest (PH) survey. The increases are largest in 2018-19 where we also observe the largest increase in the count of household wage workers. Although increasing enterprise activity is a more common reported response to flood shocks than other shocks, it still remains relatively uncommon. In addition to null effects of flood exposure on the count of household members engaged in NFE, we find no effect on the likelihood a household has any NFE and on reported total hours of work in NFE in the last 7 days during the PP survey, though hours during the PH survey increase by 10%.

Table 4: Impacts of survey-reported 2012 community flood exposure on labor supply to different activities

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Any HH farm activity	11691	0.69 [0.46]	0.02* (0.01)	0.02 (0.01)	0.02 (0.02)	0.06 (0.05)
Any HH non-farm enterprise activity	11691	0.78 [0.41]	0.02 (0.02)	0.00 (0.02)	0.04 (0.02)	0.02 (0.04)
Any wage employment activity	11691	0.43 [0.49]	0.02 (0.02)	0.02 (0.03)	0.01 (0.02)	0.04 (0.03)
Total HH farm hours, last 7 days PP	11691	41.61 [56.75]	-4.58* (2.74)	-2.01 (2.88)	-6.81* (3.50)	-8.21* (4.28)
Total HH enterprise hours, last 7 days PP	11691	39.89 [45.89]	0.94 (2.02)	1.93 (1.95)	0.43 (2.84)	-2.23 (3.59)
Total HH wage hours, last 7 days PP	11691	24.86 [42.33]	3.70** (1.50)	3.40** (1.70)	3.63** (1.71)	5.86** (2.49)
Total HH non-farm hours, last 7 days PP	11691	64.75 [58.09]	4.64* (2.44)	5.33** (2.44)	4.06 (3.19)	3.63 (4.92)
Total HH farm hours, last 7 days PH	11691	33.30 [46.01]	-2.49 (2.43)	-4.88* (2.89)	-0.51 (2.77)	1.30 (4.24)
Total HH enterprise hours, last 7 days PH	11691	38.10 [45.80]	3.74** (1.64)	3.80** (1.87)	4.68*** (1.72)	-1.68 (3.96)
Total HH wage hours, last 7 days PH	11691	27.05 [41.38]	1.59 (1.57)	2.04 (1.68)	0.90 (1.70)	2.58 (2.36)
Total HH non-farm hours, last 7 days PH	11691	65.15 [55.92]	5.33*** (2.06)	5.84*** (2.23)	5.58** (2.19)	0.90 (5.03)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. See [Online Appendix C](#) for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Despite the intensive margin increase in non-farm labor supply, we find no significant effects on non-farm earnings ([Table 2](#)). Point estimates for NFE earnings are negative and statistically significant in 2015-16. This could reflect damages from floods or reduced local consumption, though in this case we would have expected larger decreases in earnings in 2012-13. Point estimates for effects on wage earnings are positive but not significant in any round. One reason wage earnings may not increase despite increased work hours could be decreases in wages following increased

supply of workers in a context where economic damage may have decreased labor demand.²⁶ We do not find consistent effects of flood exposure on wages (Table A15). Reported wages earned by household wage workers during the PP period – when labor hours increase – are 24% lower on average after the floods, but average effects are positive and non-significant for PH wages and for wages paid to hired agricultural laborers. The point estimates for the effect on average daily wages for men’s agricultural labor reported in the community survey are negative but not significant.

We also find no evidence of households exiting agriculture. If anything, households in flooded communities are 2 pp (3%) *more* likely to engage in crop or livestock production on average after flood exposure, though this is only marginally statistically significant on average and not significant in any round. The average effect on the count of household members working in household farm activities is not statistically significant, but we find decreases in reported total hours of household farm work in the 7 days prior to the PP survey in 2015-16 and 2018-19 and in the 7 days prior to the PH survey in 2012-13. This could suggest some reallocation of household labor, but the largest decreases in farm hours do not align with the largest increases in non-farm hours.

Taken together, households do increase labor supply to non-farm livelihood activities in response to floods, with aggregate non-farm hours increasing by an average of 7% in the PP period and 8% in the PH period and larger increases in 2012-13. Increases in NFE hours at PH do not persist to 2018-19 but increases in wage hours at PP are largest in that round. This suggests that increased NFE labor may be primarily a shock coping mechanism whereas increased wage employment may represent a persistent change in household livelihood strategies. Greater engagement in non-farm livelihood activities is concentrated on the intensive margin among households already active in NFE or wage employment, and does not come at the expense of engagement in household agriculture. Farm work hours in the PP period do fall by 11% on average after the 2012 floods, which may help explain the fall in the value of farm production. Increased non-farm work hours do not, however, increase non-farm earnings so household do not diversify or significantly change their earnings shares from different activities. Estimated effects on total non-farm income are very noisy and not significant on average or in any round, but the point estimate is also positive only in 2012-13 and is negative in the following two rounds (Table A13). The results for wage hours and earnings during the PP period are consistent with constraints to households in increasing wage employment income, despite the persistent increase in wage labor engagement.

An alternative livelihood response to a flood shock could be (internal or external) migration of household members, which could generate remittance income through work not reflected in the household survey as this does not cover members who have left the household to work. We do not observe data on total remittances separately in the GHSP, but total income from assets, investments, pensions, remittances and assistance/relief increase by 145 USD (38%) average on following the 2012 floods, with the largest increases in 2012-13 and 2015-16 and no effect in 2018-19 (Table A13). This effect could reflect receipt of assistance after flood exposure or sale of assets,

²⁶Mueller and Osgood (2009b) find that droughts lead to persistent decreases in rural wages in areas more dependent on agriculture.

both reported as common flood shock coping strategies (Figure 3), as well as remittances from migrated household members.

Prior studies have shown mixed effects of flood exposure on short-term out-migration. Migration for work is not a frequently reported coping strategy in this context. Few households report any member migrating to find work in response to a shock, in line with null average and immediate effects we estimate in Table 2 and in a study of the impacts of temperature shocks on migration in Northern Nigeria (Dillon et al., 2011). We also find no average effects of flood exposure on the likelihood a household member leaves for any reason, on the likelihood any adult joins the household, on the count of household members, and on the count of young children (Table A14). These results indicate that individual migration for work is not a common short-term response to floods in Nigeria, but the positive and significant effect in 2018-19 suggest that this could be a result of individual and household migration constraints, which may be greater in the short-term following flood exposure.

5.4 Explaining crop production value effects

The effect of the 2012 floods on the value of household farm production is driven by a persistent large decrease in the value of crop production. Many studies report that floods can decrease agricultural output, with a focus on direct damages. Crop production value falls by 31% in 2012-2013, consistent with direct damage and destruction of crops by floodwaters. Indeed, such direct effects on crops are an important source of the household flood reports we use to define the flood exposure measure. Effects in subsequent survey rounds – a significant 29% decrease in 2015-16 and a noisy 26% decrease in 2018-19 ($p=0.127$) – imply that there are also persistent *indirect* effects of flood exposure on crop production. On average, 31% of crop production is sold but this does not change after flood exposure indicating households are not changing marketing or consumption decision for their crop output. Decreases in the value of crop sales are generally proportional to the decrease in the overall value of crop production.

Decreased production value could be explained by reallocation of household labor away from crop production, decreases in other crop production inputs, decreases in land productivity, or decreases in crop sales prices which we use to estimate total production value. Table 5 presents estimates of the impact of flood exposures on household crop production outcomes to test these potential explanations across all households, including those not cultivating crops.

We find no average effect on the number of crops produced, but the point estimate is negative and the HHI for area planted across crops increases significantly indicating a greater concentration of crop area. The fall in the number of crops is significant in 2015-16 while the HHI only increases significantly in 2018-19, suggesting marginally significant effects on planting decisions both across and within crops. Total area planted across crops is unchanged though the point estimates are negative. Despite negative effects of flood exposure on household farm hours in the last 7 days at the time of the PP and PH surveys and the negative point estimate for area planted, we find no effects on total family crop labor days and hours aggregated over the whole year. The point

estimates for household crop labor are in no case close to being statistically significant but are positive on average – though negative in 2018-19 – indicating no reallocation of labor away from crop production. On the other hand, flood exposure decreases days of hired or exchange crop labor by 66% on average, with significant decreases in both 2012-13 and 2018-19. While households do not significantly change their use of inorganic fertilizer or total spending on crop production inputs on average, the point estimates are negative and we find a significant reduction in fertilizer use in 2012-13. In aggregate, effects of flood exposure on crop production decisions in subsequent years appear minimal but the pattern of results suggests some potential substitution of paid inputs with household labor.

Table 5: Impacts of survey-reported 2012 community flood exposure on household crop production outcomes

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Number of crops cultivated	11691	1.93 [1.91]	-0.07 (0.08)	-0.01 (0.09)	-0.17* (0.10)	0.07 (0.18)
Household crop area planted HHI	6216	0.55 [0.26]	0.03* (0.02)	0.02 (0.02)	0.04 (0.02)	0.08** (0.04)
Total area planted (ha)	11691	1.37 [2.84]	-0.06 (0.17)	-0.04 (0.19)	-0.07 (0.17)	-0.12 (0.16)
Total family crop labor days in past year	11691	121.81 [345.11]	9.27 (24.87)	21.05 (32.98)	2.90 (26.28)	-27.64 (37.01)
Total family crop labor hours in past year	11691	956.05 [1971.47]	112.01 (121.36)	176.75 (150.85)	66.75 (158.76)	-35.91 (242.81)
Total hired/exch. crop labor days in past year	11691	7.73 [22.19]	-5.08* (2.75)	-4.96* (2.56)	-3.22 (3.68)	-15.76** (7.74)
Any inorganic fertilizer use	11691	0.33 [0.47]	-0.02 (0.02)	-0.04* (0.02)	0.01 (0.02)	-0.06 (0.06)
Total spending on crop production (USD 100s)	11691	2.80 [7.95]	-0.65 (0.49)	-0.23 (0.58)	-0.91 (0.61)	-1.72 (1.07)
HH crop yield index (normalized)	5889	0.06 [1.00]	-0.18* (0.09)	-0.25** (0.11)	-0.12 (0.10)	-0.03 (0.15)
HH crop sales price index (normalized)	5889	0.00 [1.18]	0.05 (0.07)	0.04 (0.07)	0.04 (0.10)	0.19* (0.12)
Total crop production value (USD 100s)	11691	15.18 [32.69]	-4.50*** (1.50)	-4.68*** (1.63)	-4.40** (1.81)	-3.91 (2.55)
Total crop sales (USD 100s)	11691	4.81 [13.98]	-1.45** (0.59)	-1.01 (0.72)	-1.93*** (0.74)	-1.51** (0.65)
Proportion of value of crop production sold	6216	0.31 [0.35]	-0.01 (0.02)	0.01 (0.03)	-0.03 (0.03)	0.01 (0.04)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Online Appendix C](#) for details on variable construction. The sample includes all households regardless of engagement in crop production. Households not growing crops are assigned values of 0 for most outcomes. Sample sizes are smaller for the crop area HHI and proportion of crop value sold which are only defined for households with non-0 area planted or crop production, and for the crop yield and sales price indices which are only defined for households producing one of the eight most commonly grown crops in Nigeria. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Consistent with crop destruction, flood exposure has a large and immediate effect on physical crop yields. We normalize yields and sales by crop within each round for the eight most commonly

cultivate crops in the sample²⁷ and then calculate area-weighted means across crops produced by each household and normalize these indices within round. Crop yields fall by 0.18 standard deviations (SDs) on average post-flooding, though the effect does not persist after the 0.25 SD decrease in 2012-13 which can be attributed to direct crop damages due to floods. Looking at specific commonly-cultivated crops in Nigeria, we find consistent negative point estimates and large and significant average decreases in yields for rice, cowpea, and groundnut (Table A16). In contrast with the pattern for the yield index, for some crops the largest magnitude effects are not in 2012-13, and we find significant decreases in cowpea yields in 2015-16 and in maize and rice yields in 2018-19. Over the long-term, flooding could decrease productivity through soil erosion, siltation, or contamination, and increased salinity (in coastal areas), or increase it by depositing nutrients and sediments. Overall, these results indicate no consistent persistent effect of flood exposure on physical crop yields, but that some lingering adverse effects could explain part of the persistent decrease in crop production value.

Crop sales prices would generally be expected to rise in the short-term following flood exposure due to destruction of local output and potential supply disruptions. We find no significant effects on the crop sales price index, though this increases significantly in 2018-19 where we also find a significant increase in sorghum sales prices. Rice sales prices decrease on average but there are no other effects of flood exposure, and point estimates are in general not consistently positive. Turning to market price data collected in the community surveys, we find no significant effects of flood exposure on prices of common food goods including shelled maize, local rice, and yam roots, though bread prices fell significantly (Table A15). Flood-related agricultural destruction therefore did not differentially affect prices in flood-exposed communities, indicating a strong level of market integration.

We conduct a descriptive decomposition of the relative importance of yield and price changes by estimating impacts of flood exposure on crop production value controlling for potential mediators, among households cultivating any of the eight most commonly-cultivated crops in the sample (Table 6). Among this sample, flood exposure decreases the value of crop production by 27%. Controlling for the crop yield index reduces this to a 20% decrease while controlling for crop production decisions and the crop sales price index reduces it to 24% decrease. Controlling for all three mediators reduces the effect to a non-significant 15% decrease ($p=0.118$). These results indicate that effects of flooding on physical yield, including potential effects on land productivity, explain a larger share of the decrease in crop production value but that effects on crop production decisions are also collectively significant. Together, these factors explain 44% of the decrease.

The remaining negative effect, though not statistically significant, may be due to widespread effects on crop sales prices, which would have profited producers in unexposed communities relative to communities suffering yield reductions. Looking at changes in mean cereal sales prices over time, we find that prices increased from 0.54 USD/kg on average in 2010-11 to 1.16 USD/kg in 2012-13, consistent with widespread supply disruptions, before falling back to 0.86-0.88 USD in 2015-16

²⁷These are maize, sorghum, millet, rice, cowpea, groundnut, yam, and cassava.

Table 6: Impacts of survey-reported 2012 community flood exposure on household crop production value among main crop producers

	(1)	(2)	(3)	(4)
	Total crop production value (USD 100s)			
Exposed to 2012 flood shock (recentered)	-7.88*** (3.01)	-5.85** (2.96)	-6.84** (2.90)	-4.42 (2.82)
HH crop yield index (normalized)		11.49*** (0.96)		12.74*** (0.98)
Total area planted (ha)			1.26*** (0.35)	2.49*** (0.38)
Number of crops cultivated			4.00*** (0.59)	3.78*** (0.54)
Household crop area planted HHI			-2.50 (3.04)	-3.90 (2.76)
Total hired/exch. crop labor days in past year			0.05*** (0.02)	0.04*** (0.01)
Any inorganic fertilizer use			2.71 (1.75)	2.35 (1.70)
Total spending on crop production (USD 100s)			0.11 (0.08)	0.09 (0.08)
HH crop sales price index (normalized)			-0.71 (0.55)	0.03 (0.50)
Observations	5889	5889	5889	5889
Baseline mean, non-flooded	28.84	28.84	28.84	28.84

Note: This table presents average effects of 2012 community flood exposure on total crop production value in 2016 USD PPP in subsequent survey rounds, following Table A16. The sample includes all households that cultivated any of the eight main crops cultivated in the sample. The first column is the main specification for the impacts of flood exposure restricted to this sample. The next three columns introduce controls for crop yields, crop sales prices, and crop production decisions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

and 2018-19. Again consistent with strong market integration, these price changes did not differ by community flood exposure. Higher crop sale prices in unexposed communities would amplify differences in the value of crop production. Table A9 shows that the estimated effects of floods on crop production value are if anything larger if we exclude nearby communities and those in the same LGA from the sample of controls, suggesting limited local spillovers. But widespread crop sale price increases together with decreased production in exposed communities could explain part of the residual decrease in crop production value in flood-exposed communities after controlling for impacts on yields and crop production decisions.

5.5 Heterogeneity in responses to flood exposure

We next examine heterogeneity in the effects of flood exposure to better understand which households are more vulnerable to flood shocks and how adjustment occurs across livelihood activities. Specifically, we estimate versions of our baseline specification that allow the treatment effect to vary across baseline household and community characteristics, including geography, market access, prior flood exposure, and household size, assets, and livelihood activities. We focus on effects of flood exposure on consumption and on production value or earnings and member engagement in the three types of livelihood activities we consider. In each regression, we estimate a triple-differences model fully interacting a baseline household or community characteristic H with the Post and

Flood dummies and the state-by-round fixed effects (following Feigenberg et al., 2025).

Across most dimensions, we do not find consistent or statistically distinguishable differences in treatment effects across outcomes. In particular, heterogeneity by baseline household size, asset holdings, proximity to a major road, market, or town, proximity to water, fluvial flood risk, and recent community flood exposure yields mixed or imprecise patterns (Table A17). While some of these characteristics are associated with differences in wage or NFE earnings, they do not consistently predict differential adjustment in labor allocation or earnings outcomes. This suggests that vulnerability to flood shocks does not map cleanly into a broad ranking based on demographics, wealth, community centrality, or general geographic exposure. One exception is that households in flood-exposed communities that are closer to major markets, roads, or towns consistently experience larger decreases in NFE earnings following the floods, likely due to greater engagement in these vulnerable activities in more urban areas.

The only dimension along which there appears to be informative heterogeneity is baseline livelihood activities (Table 7). As would be expected mechanically, Panel A shows that decreases in the value of agricultural production are concentrated among households engaged in agriculture at baseline, though the difference is not statistically significant suggesting there are also worse outcomes for new entrants into agriculture in flooded communities. But baseline agricultural households are not significantly worse off in terms of consumption than non-agricultural households. They also increase the count of members engaged in wage work by significantly more, with no increase for non-agricultural households. Somewhat less intuitively, Panel B shows that the fall in agricultural production is also only significant in households engaged in wage work at baseline, though again the difference is not significant. But flooding does not decrease consumption for these households, and the results suggest this may be because they respond with much larger labor allocations to wage work ($p = 0.01$), while also earning significantly more wage income ($p = 0.10$) to offset the decrease in farm production value. Panel C shows that the strongest difference in effects on the value of farm production is by baseline engagement in NFE, with the decrease driven entirely by households with any NFE at baseline ($p=0.01$). As with panels A and B, households with larger decreases in farm production value do not have larger decreases in consumption, and in fact the opposite is true ($p=0.12$), despite the fact that households not engaged in NFE significantly increase the count of wage workers and increase wage earnings by significantly more ($p=0.02$).

Panel D shows that the patterns in panels A and B are driven by the 28% of households engaged in *both* agriculture and wage work at baseline. These households drive the effects on farm production value ($p=0.12$ for the difference) and the count of wage workers ($p=0.02$) and do not experience a significant decrease in consumption following flood exposure. Panel E further shows an even larger difference in the effect of flood exposure on farm production for households engaged in both agriculture and NFE at baseline ($p < 0.01$), though outcomes for these households are not otherwise significantly different. We note that these tests do not have any causal interpretation, as differences in baseline household livelihood activities are associated with different community and household characteristics. For example, households engaged in both agriculture and NFE or both

Table 7: Impacts of survey-reported 2012 community flood exposure by baseline household activities

		HH agriculture		Wage work		HH NFE	
	(1) Daily HH cons./cap.	(2) Value prod.	(3) Count mems.	(4) Total earn.	(5) Count mems.	(6) Total earn.	(7) Count mems.
A. H = Any baseline HH agriculture (75%)							
Flood $\times$ Post, H=0	-0.77 (0.52)	-1.22 (1.34)	0.05 (0.06)	25.03 (21.75)	-0.03 (0.08)	-4.69 (10.39)	-0.02 (0.09)
Flood $\times$ Post, H=1	-0.58** (0.24)	-4.20* (2.23)	0.08 (0.12)	0.82 (4.26)	0.16** (0.07)	-1.01 (2.80)	0.02 (0.06)
p , equality of effects	0.73	0.27	0.82	0.28	0.06	0.73	0.72
B. H = Any baseline wage work (42%)							
Flood $\times$ Post, H=0	-0.74** (0.33)	-1.06 (2.43)	0.04 (0.11)	-2.33 (1.65)	0.02 (0.03)	-6.64* (3.74)	0.07 (0.06)
Flood $\times$ Post, H=1	-0.47 (0.29)	-4.64** (2.28)	0.12 (0.12)	18.05 (12.29)	0.31*** (0.11)	-1.17 (5.07)	0.04 (0.08)
p , equality of effects	0.54	0.27	0.59	0.10	0.01	0.38	0.79
C. H = Any baseline NFE (77%)							
Flood $\times$ Post, H=0	-1.07** (0.43)	4.86 (3.52)	-0.13 (0.13)	20.52*** (7.14)	0.22*** (0.08)	-0.68 (1.31)	-0.07 (0.07)
Flood $\times$ Post, H=1	-0.36* (0.20)	-4.94*** (1.53)	0.08 (0.08)	0.26 (5.67)	0.10 (0.06)	-4.52 (3.65)	0.04 (0.06)
p , equality of effects	0.12	0.01	0.11	0.02	0.20	0.32	0.19
D. H = Both HH agriculture and wage work at baseline (28%)							
Flood $\times$ Post, H=0	-0.70** (0.29)	-0.78 (2.14)	0.03 (0.09)	4.27 (5.95)	-0.00 (0.04)	-5.98 (3.85)	0.07 (0.06)
Flood $\times$ Post, H=1	-0.32 (0.25)	-6.65** (3.20)	0.14 (0.18)	7.56 (10.99)	0.31** (0.13)	3.34 (5.19)	0.03 (0.10)
p , equality of effects	0.31	0.12	0.53	0.79	0.02	0.14	0.73
E. H = Both HH agriculture and NFE at baseline (57%)							
Flood $\times$ Post, H=0	-0.85** (0.35)	3.04 (2.61)	-0.01 (0.09)	10.56 (10.70)	0.11 (0.07)	-5.05 (5.52)	-0.02 (0.07)
Flood $\times$ Post, H=1	-0.45* (0.24)	-6.98*** (2.10)	0.10 (0.12)	0.08 (5.26)	0.14* (0.08)	-1.18 (3.54)	0.04 (0.07)
p , equality of effects	0.32	0.00	0.37	0.38	0.75	0.55	0.54

Note: This table presents average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds, by baseline household characteristics H as indicated in the panel headings. For each H , we show the share of households for which $H = 1$. Each column represents an outcome. Monetary values are in 2016 PPP USD. See [Online Appendix C](#) for details on variable construction. The results for each column in each panel are from a single triple-differences regression fully interacting the baseline household characteristic H with the Post and Flood dummies and state-by-round fixed effects. We calculate the average effect in the group with $H = 1$ by taking the sum of the $Flood \times Post$ and $Flood \times Post \times H$ coefficients and estimate SEs using the *xtlincom* function in Stata. We also report the p -value for the triple interaction term, which tests for equality of effects between the two groups. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

agriculture and wage work typically have more members, more assets, and are more likely to be urban and nearer to major roads, markets, and towns ([Table A18](#)). Household fixed effects may absorb some of these differences, but in general the interpretation of differences in [Table 7](#) can only be seen as suggestive of correlations.

The patterns of flood exposure effects by baseline livelihood activities imply that households with diversified livelihood activities at baseline were more vulnerable to flood damages in agriculture.

They also indicate that decreases in farm production value may not drive decreases in consumption. Although the differences are never statistically significant, decreases in daily consumption per capita are consistently larger in magnitude for the households experiences non-significant effects of floods on farm production. More generally, effects on consumption do not align with effects on total household earnings and production value. This could reflect challenges in estimating consumption expenditures for goods the household produces itself – potentially explaining weaker consumption effects among agricultural households – or effects on market prices, though we do not find evidence of price increases following the floods (Table A15).

The results also re-emphasize that labor supply impacts are driven on the intensive margin, and that this response is largely concentrated in wage work. The increase in the count of wage workers in particular driven entirely by households that already had members in wage employment at baseline. But these households do not report earning significantly more wage income. This suggests that labor supply adjustments along existing margins may be incomplete or low-productivity, limiting contributions to household resilience and recovery. Why don't more households respond to the flood shock by participating more in wage work? One possibility is that agricultural households – those experiencing the largest income losses due to flood exposure – live in areas with limited opportunities for wage work. But effects on the count of wage workers are significantly larger in *less* urban areas, which also experience a larger increase in the count of members active in NFE (Table A17). This pattern could imply that these increases are coping strategies following farm production shocks, but we do not find consistently larger decreases in farm production value in more rural areas.

Taken together, these patterns suggest limited scope for household adjustment along new margins following large shocks. While households appear able to reallocate labor within existing activities, they do not readily enter new activities in response to flood exposure. This would be consistent with the presence of short-run barriers to occupational switching, such as liquidity constraints, skill requirements, or limited local demand for labor. The fact that wage work responses are not larger for households in communities where we would expect larger labor demand suggests the main constraints to livelihood responses are at the household level. These findings help explain why we do not observe widespread shifts in livelihood structure in response to flooding: adjustment occurs primarily along margins that are already available to households.

6 Conclusion

Floods are among the most common and destructive natural disasters, and climate change is increasing flood risk. This has implications for the well-being and livelihoods of households in areas at risk of flooding. Increased flood risk in agricultural communities may affect household labor market decisions and livelihood strategies by changing both expected returns to different activities and resources available to invest in them.

We find that, controlling for predicted risk of flooding, survey-reported community-level expo-

sure to floods during Nigeria’s 2012 flood disaster decreases crop production and household food consumption, with some persistence over the seven years after exposure. Consistent with broader concerns that climate change may limit structural transformation in vulnerable rural economies (e.g., Cruz, 2024; Nath, 2025), we find no impacts of flooding on agricultural exit despite decreases in agricultural productivity in flood-exposed communities. These results suggest that severe flood shocks may reinforce existing livelihood patterns rather than induce diversification away from agriculture in this setting.

As in many studies showing short-term increases in wage employment following agricultural shocks, we find evidence of persistent increases in household labor allocations to wage work. However, this does not represent livelihood diversification. The absence of corresponding increases in wage earnings suggests limits to the returns from expanding wage labor supply following flood exposure. Delayed increases in migration may reflect attempts to cope with persistent consumption decreases in settings with limited local non-farm opportunities. Future work analyzing constraints to and drivers of livelihood responses to shocks over time, building on broader work on climate adaptation in sub-Saharan Africa (Di Falco et al., 2012; Gbetibouo et al., 2010), is needed to support greater resilience to flooding.

A key result of this study is that how flood exposure is measured determines what we conclude about impacts on non-farm livelihoods. While recent work has increasingly emphasized satellite-based flood measures, our results highlight the value of survey reports as measures of economically harmful flood exposure. Although survey reports will not capture all physical inundation, they likely reflect flooding most associated with household coping and adaptation responses. NASA’s flood maps based on satellite imagery detect visible surface water anomalies but may be poorly aligned with measures of household-relevant flood damage in settings with significant and persistent cloud cover, pluvial flooding, and small-scale inundation. Combining multiple data sources, including radar-based remote sensing and survey reports, may improve future measurement of flood exposure. Researchers must therefore carefully consider both definition and measurement issues when estimating and interpreting economic impacts of flood exposure.

One limitation of the study is that we study a single flood event. If many households have already adapted to increased flood risk the analysis may fail to capture longer-run effects of changes in flood risk and resulting adaptation responses of households. This concern is mitigated in this study given the extreme severity of Nigeria’s 2012 floods and we find no consistent differences in impacts of exposure by community flooding over the preceding five years, but future work on the long-term economic impacts of flood exposure should consider effects of changes in flood hazard and cumulative flood exposure over longer time frames, as in recent work by Patel (2024) and Sajid (2023).

The results are also relevant to policymakers in developing countries. The inability of this MODIS-based flood product to identify many communities reporting flood damages has implications for emerging recommendations to use satellite data for policy targeting. For disaster response and social protection, remotely-sensed flood maps may miss communities experiencing economically

damaging pluvial or small-scale floods and allocate resources differently than approaches based on local reports of flood impacts. For research and index insurance, validation against household- or community-level damage reports should be a core component of the design. More broadly, the results highlight both the challenges households face in adapting to severe flood shocks and the importance of carefully measuring flood exposure when designing policies and evaluating climate adaptation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work we used ChatGPT Plus in order to check the code for errors, help write the code for the analysis of satellite data, and provide suggestions for editing the paper. We used several AI paper review tools to generate suggestions for improving the paper, including paperreview.ai, Refine.ink, and Coarse.ink. After using these tools, we reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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A Additional Figures

Figure A1: GHSP data collection timeline

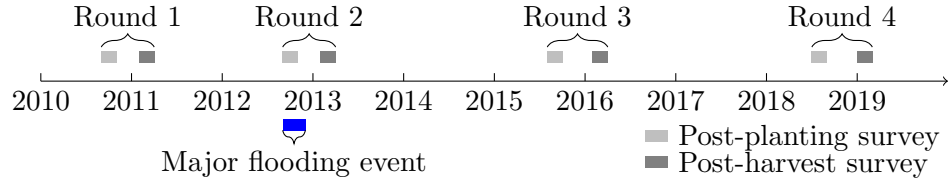
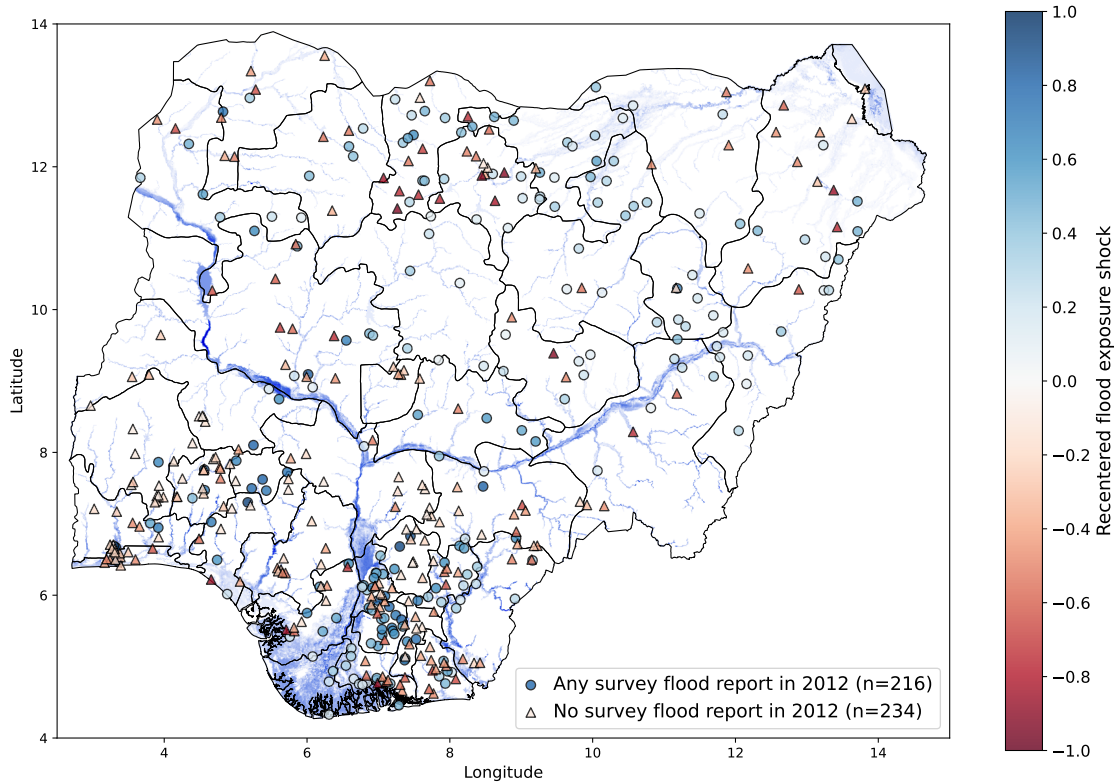
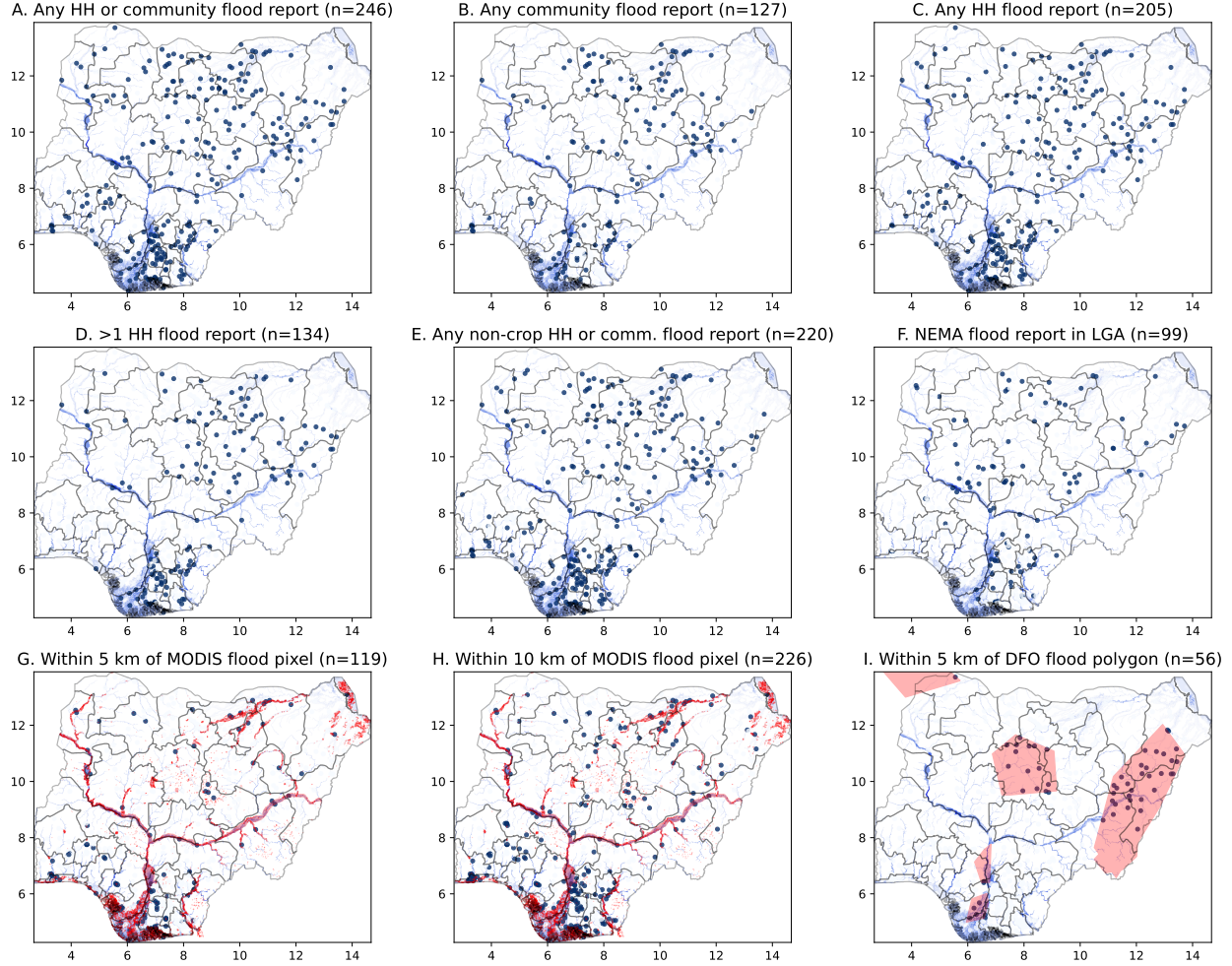


Figure A2: Recentered survey-reported 2012 community flood exposure, accounting for predicted 2012 exposure



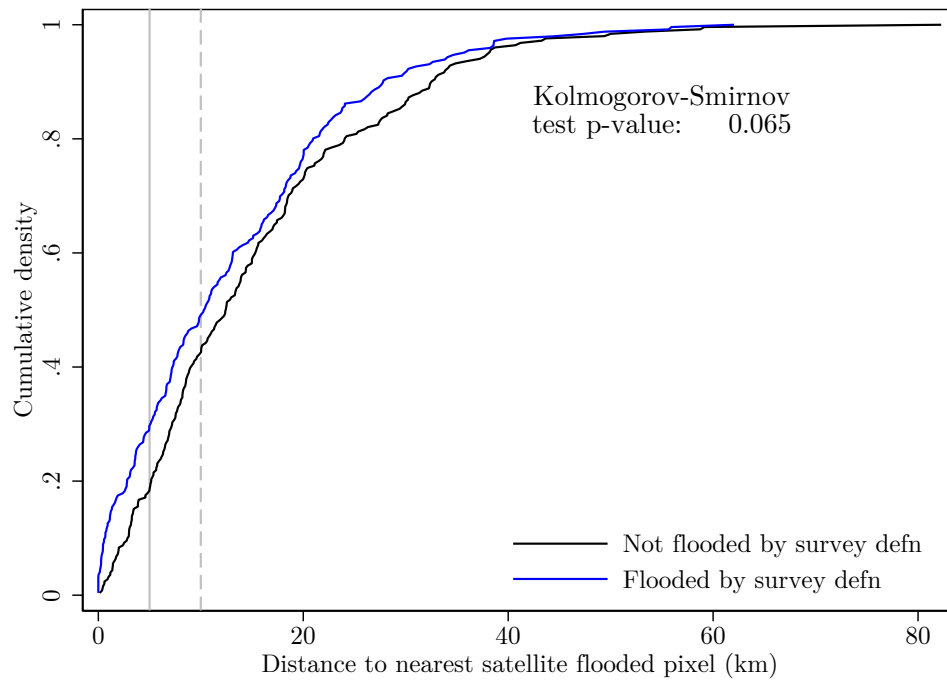
Note: The figure plots the locations of GHSP communities in the analysis sample by survey-reported 2012 community flood exposure and recentered flood treatment. The smaller number of flooded communities relative to [Figure A3 Panel A](#) is a result of balancing on recent flood history, underlying flood hazard, and predicted 2012 flood exposure. Flood exposure is defined as any household or community report of negative effects of floods within a community in 2012. The recentered flood treatment subtracts the estimated 2012 flood propensity from the binary exposure. A value of 1 indicates a flooded community with an estimated flood propensity of 0, and a value of -1 indicates a non-flooded community with an estimated flood propensity of 1. The blue shading in the background corresponds to the depth of 100-year fluvial floods.

Figure A3: Community flood detection by flooding measure



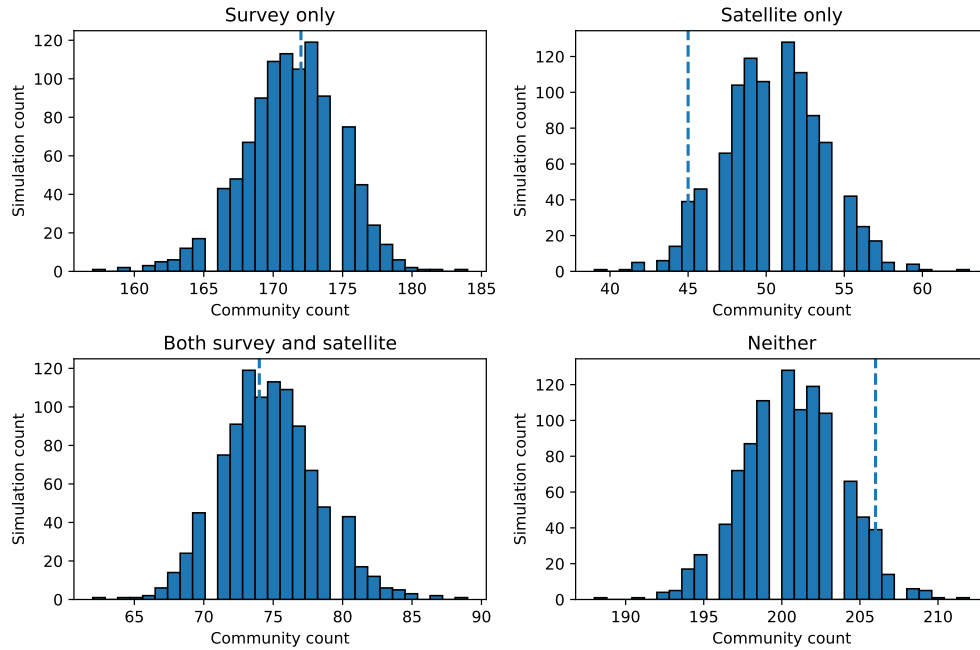
Note: Each panel shows the set of GHSP communities identified as exposed to floods in 2012 according to a different definition of flood exposure. The blue shading in the background of each panel corresponds to the depth of 100-year fluvial floods. Panels A-E rely on survey reports of floods in 2012. Panel A shows communities with any household or community survey flood report. Panel B shows communities with any community informant flood reports. Panel C shows communities with any household reports of floods in any module. Panel D shows communities with at least two such household reports. Panel E shows communities with any household non-crop-related flood or community flood report. Panel F shows communities in local government areas (LGAs) where NEMA reports any floods in 2012. Panels G and H show communities within 5 and 10 km of any pixel identified as flooded by MODIS satellite imagery at any point in 2012 in the NASA Global Flood Product. Panel I shows communities within 5 km of polygons where flooding is reported in the Dartmouth Flood Observatory archive. The flooded pixels and polygons in panels G-I are colored red.

Figure A4: Distance to nearest flooded pixel by survey flood definition



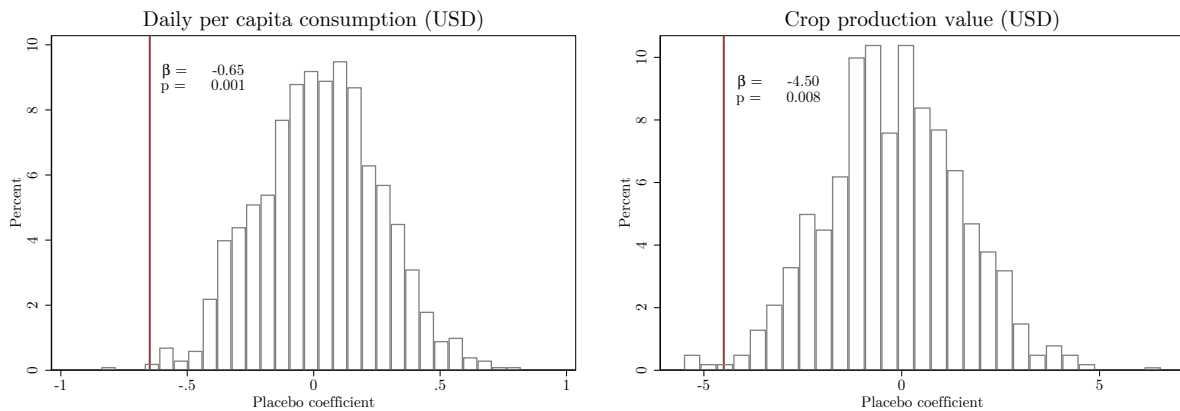
Note: The figure plots the cumulative distribution of GHSP community distance to the nearest pixel identified as flooded at any point in 2012 in the NASA MODIS flood database by whether the community was exposed to the 2012 floods according to our main survey-based definition. For each community, we calculate straight-line distances between the jittered community centroid and the centroids of 250 m MODIS flooded pixels. The solid vertical line represents the 5 km cutoff used in the main satellite-based flood exposure definition, and the dashed vertical line represents a 10 km cutoff. We also show the p -value for the Kolmogorov-Smirnov test of equality of the two distributions.

Figure A5: Flood exposure classifications under 1000 simulations of community coordinate displacement



Note: The figure shows the distribution of overlap classifications between survey-reported and satellite-detected flood exposure across 1,000 simulations of LSMS-style coordinate displacement. Urban communities are randomly displaced up to 2 km and rural communities up to 5 km, with a 1% probability of displacement up to 10 km for rural communities. Satellite exposure is defined as being within 5 km of a MODIS-detected flood. Dashed vertical lines indicate classifications using the publicly released community coordinates. The limited variation in overlap counts across simulations suggests that coordinate displacement contributes only modestly to the disagreement between survey and satellite measures of flood exposure.

Figure A6: Distribution of impacts of permuted 2012 survey-reported flood exposure within flood propensity bins



Note: The two panels show the results of 1000 simulations of the impacts of 2012 recentered flood exposure permutating actual exposure within bins of estimated flood propensity to keep the same number of flooded communities in each bin. For each outcome, we plot the distribution of the estimated coefficients from each simulation. The vertical red line shows the coefficient for the effect of actual realized flood exposure. The text alongside the red line shows the coefficient estimate and the randomization p -value, calculated as the share of simulation estimates more extreme (in absolute terms) than the observed estimate. A small p -value indicates a very low likelihood that we would observe our estimate under the null of random allocation of flood exposure within flood propensity bins.

B Additional Tables

Table A1: Predictors of 2012 flood exposure

	(1) Any HH or comm flood rept	(2) Any HH flood rept	(3) >1 HH flood rept	(4) HH non-crop or comm. flood rept	(5) <5 km from MODIS flood	(6) <10 km from MODIS flood
Distance to nearest water area (km)	-0.012** (0.005)	-0.015** (0.006)	-0.023*** (0.007)	-0.003 (0.004)	-0.020** (0.009)	-0.031*** (0.006)
Avg 100-year flood depth within 5 km	0.508** (0.216)	0.750** (0.315)	0.623*** (0.192)	0.467*** (0.165)	2.579*** (0.649)	3.190*** (0.802)
Area equipped for irrigation in cell in 2005 (100 ha)	0.008 (0.007)	0.009 (0.007)	0.012* (0.007)	0.012** (0.006)	0.005 (0.007)	0.009 (0.008)
Percent ag land w/in approx 1 km	0.007 (0.006)	0.013* (0.008)	0.003 (0.006)	0.007 (0.004)	-0.020 (0.013)	-0.017** (0.007)
Sparse vegetation (>15%)	0.000 (.)	1.963 (1.390)	0.000 (.)	1.114 (1.246)	2.128 (1.765)	0.973 (1.635)
>50% artificial surfaces and associated urban areas	-0.609 (0.489)	-0.619 (0.625)	-1.211 (0.858)		0.078 (0.862)	0.824 (0.562)
Platform terrain (very low plateaus)	-0.692* (0.395)	-0.320 (0.499)		-0.311 (0.356)		0.927** (0.384)
Low plateaus terrain		-1.883* (1.055)	1.299* (0.777)		-1.914 (1.869)	1.011 (0.810)
Slope (percent)	-0.013 (0.061)	-0.020 (0.098)	-0.130* (0.073)	-0.092* (0.053)	-0.086 (0.125)	-0.064 (0.070)
Total precipitation of wettest quarter in 2012 (100 mm)	0.162* (0.084)	0.338 (0.333)	0.313** (0.127)	0.217** (0.108)		
Avg. precipitation of wettest quarter historically (mm)		-0.006** (0.003)	0.002 (0.002)	0.001 (0.001)	-0.000 (0.002)	
Potential Wetness Index	-0.037 (0.030)	-0.020 (0.035)		-0.038 (0.029)	-0.032 (0.047)	
Percent ag land X 2012 deviation from avg. annual precipitation (100 mm)	0.292 (0.200)		0.204 (0.212)			0.165 (0.410)
Percent ag land X 2012 deviation from avg. wettest qtr. precip. (100 mm)		0.004 (0.003)		0.004** (0.002)	0.021*** (0.005)	0.011* (0.007)
Elevation (100 m) X 2012 deviation from avg. annual precipitation (100 mm)	3.998 (3.650)	15.915** (6.586)	3.758 (2.629)		3.504 (6.886)	
HH distance to nearest market (km)	0.004 (0.003)	0.004 (0.004)			0.001 (0.005)	0.006* (0.004)
Any HH rept. flood shock in 2007		-0.862 (0.614)		-1.119** (0.548)	-2.195* (1.265)	-1.381** (0.682)
Any HH rept. flood shock in 2008	0.410 (0.611)	0.974 (0.696)	0.648 (0.635)	1.126** (0.555)	-0.714 (1.063)	-0.900 (0.704)
Any HH rept. flood shock in 2009	1.175*** (0.436)	1.149** (0.447)	1.422*** (0.399)	0.943*** (0.359)	0.991* (0.580)	
Any HH rept. flood shock in 2010	-0.682* (0.361)	-0.327 (0.396)			0.899* (0.525)	
Any HH rept. flood shock in 2011		0.441 (0.441)	0.099 (0.432)		0.228 (0.564)	1.224** (0.476)
Share of HHs w/ any crop activity in 2010	0.955** (0.426)	1.998*** (0.673)	1.762*** (0.630)	0.588* (0.333)	-0.763 (0.767)	
Mean HH area planted with crops in 2010 (ha)	0.055 (0.126)	0.341** (0.152)	0.283** (0.129)		0.028 (0.204)	-0.142 (0.145)
Share of HHs experiencing any food insecurity in 2010		0.768 (0.662)	1.490** (0.680)		-0.648 (0.881)	1.114* (0.637)
Observations	493	496	481	497	472	488

Note: This table shows the results of logit regressions predicting 2012 flood exposure under different measurement approaches indicated in the column headings. See [Figure A3](#) for a description of the different measures. The predictors are selected from a broader set of geographic, weather, and mean household characteristics in a first stage using a Lasso regression with 10-fold cross-validation. Differences in the number of community observations result from certain selected land cover classes predicting failure or success perfectly, and observations in those land cover classes dropping out from the estimation. For simplicity, we only show characteristics selected for at least four flood measures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Estimated average impacts of 2012 flood exposure by flood measure

	Any HH or comm flood rept	Any HH flood rept	>1 HH flood rept	HH non-crop or comm. flood rept	<5 km from MODIS flood	<10 km from MODIS flood
	(1)	(2)	(3)	(4)	(5)	(6)
Any food insecurity in last 12 months	-0.02 (0.02)	0.00 (0.02)	0.04 (0.02)	-0.01 (0.02)	0.02 (0.03)	-0.01 (0.02)
Daily HH consumption per capita (USD)	-0.65*** (0.22)	-0.78*** (0.26)	-0.61** (0.25)	-0.46* (0.24)	0.20 (0.33)	-0.29 (0.25)
Total value of HH farm production (USD 100s)	-3.13* (1.74)	-2.62 (1.85)	-4.03** (1.75)	-2.96 (2.36)	-3.78** (1.92)	-1.53 (1.60)
Total non-farm HH enterprise earnings (USD 100s)	-4.00 (3.01)	-2.88 (3.50)	2.94 (3.44)	-1.28 (3.01)	12.14*** (4.63)	0.92 (3.92)
Total wage employment earnings (USD 100s)	4.69 (5.39)	-1.30 (5.53)	1.51 (6.16)	4.31 (5.97)	-1.05 (8.86)	9.08 (6.44)
HH farm share of total HH earnings and prod value	-0.00 (0.01)	-0.01 (0.02)	-0.01 (0.02)	0.00 (0.01)	-0.02 (0.02)	-0.01 (0.02)
NFE share of total HH earnings and prod value	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)	0.01 (0.02)	0.03 (0.02)	0.00 (0.02)
Wage share of total HH earnings and prod value	-0.02 (0.01)	-0.02* (0.01)	-0.00 (0.01)	-0.02 (0.01)	-0.01 (0.02)	0.00 (0.01)
HH earnings Herfindahl-Hirschman Index	0.00 (0.01)	-0.01 (0.01)	-0.00 (0.01)	0.01 (0.01)	0.01 (0.01)	0.00 (0.01)
Count of HH members wkg. in HH farm	0.08 (0.09)	0.05 (0.09)	0.07 (0.11)	0.08 (0.08)	-0.02 (0.11)	-0.09 (0.08)
Count of HH members wkg. in HH non-farm enterprise	0.06 (0.06)	0.05 (0.05)	-0.02 (0.06)	0.05 (0.06)	0.04 (0.05)	0.05 (0.05)
Count of HH members wkg. in wage employment	0.14*** (0.05)	0.07 (0.06)	0.10 (0.06)	0.12** (0.05)	0.03 (0.07)	-0.12* (0.07)
Any member left HH for work during/since last round	-0.01 (0.01)	-0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)	-0.02 (0.01)	-0.02 (0.01)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. Each column indicates how 2012 community flood exposure is defined. For each flood measure, we estimate the 2012 flood propensity and use this to recenter flood exposure as described in [Subsection 3.3](#). Column (1) presents the results using the main survey definition of any 2012 flood report in any household survey module or in the community survey. Column (2) is similar but does not consider floods reported in the community survey, while column (3) further requires at least 2 household flood reports. Column (4) follows column (1) but excludes communities where floods are only reported in the household agriculture module or crop failure shock questions from the flood definition and from the sample. Columns (5) and (6) define communities as flooded based on proximity to pixels identified as flooded by MODIS satellite imagery from the NASA Global Flood Product, using 5 and 10 km buffers around the community centroid, respectively. All regressions include household and state-by-round fixed effects and the main analysis sample that is balanced on underlying flood hazard by 2007-2011 survey flood report strata. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Estimated average impacts of 2012 flood exposure by flood measure in the same overlapping sample

	Survey (1)	Satellite (2)
Daily HH consumption per capita (USD)	-0.63** (0.25)	0.10 (0.34)
Total value of HH farm production (USD 100s)	-2.16 (1.82)	-3.85** (1.92)
Total crop production value (USD 100s)	-3.11* (1.70)	-4.22** (1.95)
Total non-farm HH enterprise earnings (USD 100s)	-3.23 (3.44)	10.72** (4.66)
Total wage employment earnings (USD 100s)	2.80 (5.83)	-0.02 (9.40)
Count of HH members wkg. in HH farm	0.05 (0.10)	-0.02 (0.11)
Count of HH members wkg. in HH non-farm enterprise	0.01 (0.06)	0.06 (0.06)
Count of HH members wkg. in wage employment	0.12** (0.06)	0.04 (0.07)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. We consider the main survey and satellite measures of community flood exposure, and restrict the sample for both to the set of communities where both predicted 2012 flood propensities are defined. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. All regressions include household and state-by-round fixed effects and the main analysis sample that is balanced on underlying flood hazard by 2007-2011 survey flood report strata. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Balance in baseline characteristics by 2012 flood exposure

	No flood report Mean (SD)	Flood report difference (SE)	Recentered flood shock difference (SE)
<i>Community characteristics</i>			
Rural	0.62 (0.49)	0.10** (0.05)	-0.10** (0.05)
HH distance to nearest major road (km)	13.22 (17.95)	1.46 (2.21)	1.94 (1.66)
HH distance to nearest market (km)	66.72 (46.60)	6.64* (3.87)	4.05 (3.12)
HH distance to nearest pop. center w/ >20k pop. (km)	17.56 (19.29)	1.41 (2.23)	-3.28* (1.80)
Percent ag land w/in approx 1km	24.56 (26.10)	7.27*** (2.73)	1.25 (2.18)
Slope (percent)	3.33 (2.67)	-0.21 (0.27)	-0.31* (0.18)
Elevation (m)	271.00 (193.38)	-24.52 (15.96)	-17.33 (11.35)
Annual Precipitation (mm)	1431.95 (593.33)	18.82 (22.75)	-21.25 (21.05)
Distance to nearest water area (km)	37.29 (31.74)	-3.19 (2.34)	-0.18 (2.37)
Avg 100-year flood depth within 5 km	0.20 (0.44)	0.11* (0.06)	-0.07 (0.05)
Any comm. HH flood shock rept. from 2007-2011	0.19 (0.39)	0.10* (0.05)	-0.04 (0.04)
<i>Household characteristics</i>			
Female-headed HH	0.16 (0.37)	-0.01 (0.01)	-0.02 (0.02)
Count of HH members	5.24 (2.95)	0.33** (0.16)	0.06 (0.13)
Household asset index	0.07 (1.07)	-0.22*** (0.06)	0.01 (0.06)
Any food insecurity in last 12 months	0.20 (0.40)	0.02 (0.02)	0.02 (0.02)
Daily HH consumption per capita (USD)	5.84 (5.01)	-0.41 (0.26)	0.28 (0.24)
<i>Household activities</i>			
Any HH farm activity	0.64 (0.48)	0.10*** (0.03)	-0.06* (0.03)
Any HH non-farm enterprise activity	0.76 (0.43)	0.03 (0.02)	-0.00 (0.02)
Any wage employment activity	0.42 (0.49)	-0.09*** (0.03)	0.01 (0.02)
Total value of HH farm production (USD 100s)	15.36 (36.52)	6.88*** (1.71)	-1.40 (1.98)
Total non-farm HH enterprise earnings (USD 100s)	26.54 (83.20)	-5.55* (3.02)	3.28 (2.97)
Total wage employment earnings (USD 100s)	34.37 (158.96)	-17.90*** (6.19)	-4.32 (6.17)
Total area planted (ha)	1.23 (2.79)	0.92*** (0.23)	0.04 (0.19)
Test of joint significance		$F = 7.69$	$F = 1.00$
p -value		$p < 0.001$	$p = 0.462$

Note: This table shows the baseline (2010-11) control mean and difference by 2012 flood exposure status for selected community and household characteristics from a series of separate regressions. All regressions include state fixed effects, and standard errors are clustered at the community level ($N = 497$). Flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. The first column of differences uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood exposure. The bottom of the table shows results for the test of the hypothesis that the relationship between a given 2012 flood exposure measure and all variables is jointly 0. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Balance in flood risk by 2012 flood exposure

	(1)	(2)
	Any flood report	
2012 reported flood exposure	0.015 (0.010)	
2012 recentered flood exposure		-0.000 (0.011)
Observations	4205	4205
Mean, not flooded in 2012	0.042	0.042
State and Year FE	Yes	Yes

Note: This table shows the results from regressing a dummy for any community flood exposure in a given year according to the survey measure on dummies for flood exposure in 2012. The first row uses the simple binary flood treatment measure while the second recenters the treatment around predicted flood exposure. Observations are at the community-year level for 2007-2018, and 2012 is omitted. Fixed effects control for state and year. Standard errors are clustered by community, but are similar if we cluster by year. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Separating effects of flood exposure and estimated flood risk

	Flood Shock	Flood Propensity
	(1)	(2)
Any food insecurity in last 12 months	-0.02 (0.02)	0.00 (0.06)
Daily HH consumption per capita (USD)	-0.65*** (0.22)	0.13 (0.70)
Total value of HH farm production (USD 100s)	-3.13* (1.74)	-0.52 (6.68)
Total non-farm HH enterprise earnings (USD 100s)	-4.00 (3.01)	2.27 (15.09)
Total wage employment earnings (USD 100s)	4.69 (5.39)	35.99 (27.65)
HH farm share of total HH earnings and prod value	-0.00 (0.01)	-0.04 (0.04)
NFE share of total HH earnings and prod value	0.01 (0.02)	-0.01 (0.06)
Wage share of total HH earnings and prod value	-0.02 (0.01)	0.07* (0.05)
HH earnings Herfindahl-Hirschman Index	0.00 (0.01)	-0.02 (0.04)
Count of HH members wkg. in HH farm	0.08 (0.09)	0.03 (0.27)
Count of HH members wkg. in HH non-farm enterprise	0.06 (0.06)	0.11 (0.23)
Count of HH members wkg. in wage employment	0.14*** (0.05)	-0.70** (0.29)
Any member left HH for work during/since last round	-0.01 (0.01)	0.11*** (0.04)

Note: This table compares estimated effects of 2012 flood exposure following the main specification (column 1) to effects of 2012 flood propensity among non-flooded communities (column 2). The purpose of the comparison is to determine to what extent flood propensity alone may be associated with different trends over time after the major 2012 floods. In addition to dropping flooded communities from the sample, the specification in column 2 follows the main specification but replaces the recentered 2012 flood shock $\times$ post term with a 2012 estimated flood propensity $\times$ post term. Since the recentering of the flood shock in column 1 directly accounts for this flood propensity, the estimated effects control for any differential trends post-2012 by flood propensity, isolating impacts of actual flood exposure. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Sensitivity to alternative ways of controlling for non-random component of survey-reported 2012 community flood exposure

	(1)	(2)	(3)	(4)	(5)
	Recentering	No recentering			
	Main (SE)	Prediction-round controls (SE)	Risk/rain-round controls (SE)	No added controls (SE)	No added controls (SE)
Any food insecurity in last 12 months	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.02)	-0.00 (0.02)
Daily HH consumption per capita (USD)	-0.65*** (0.22)	-0.69*** (0.22)	-0.75*** (0.22)	-0.74*** (0.22)	-0.67*** (0.19)
Total value of HH farm production (USD 100s)	-3.13* (1.74)	-3.17* (1.74)	-3.04* (1.61)	-2.96* (1.62)	-3.29** (1.58)
Total non-farm HH enterprise earnings (USD 100s)	-4.00 (3.01)	-3.87 (2.98)	-2.92 (2.86)	-3.09 (2.83)	-2.55 (2.64)
Total wage employment earnings (USD 100s)	4.69 (5.39)	5.66 (5.34)	6.79 (5.00)	7.71 (5.08)	7.54 (4.65)
HH farm share of total HH earnings and prod value	-0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.00 (0.01)
NFE share of total HH earnings and prod value	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)	0.01 (0.01)
Wage share of total HH earnings and prod value	-0.02 (0.01)	-0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
HH earnings Herfindahl-Hirschman Index	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)
Count of HH members wkg. in HH farm	0.08 (0.09)	0.08 (0.09)	0.07 (0.09)	0.07 (0.09)	0.10 (0.08)
Count of HH members wkg. in HH non-farm enterprise	0.06 (0.06)	0.06 (0.06)	0.05 (0.06)	0.05 (0.06)	0.01 (0.05)
Count of HH members wkg. in wage employment	0.14*** (0.05)	0.13** (0.05)	0.07 (0.05)	0.07 (0.05)	0.09* (0.05)
Any member left HH for work during/since last round	-0.01 (0.01)	-0.01 (0.01)	-0.00 (0.01)	-0.00 (0.01)	0.01 (0.01)
Sample	Main	Main	Main	Main	All EAs

Note: This table repeats [Table A2](#) column 1 but varies how we control for the non-random component of survey-reported 2012 community flood exposure. Each column indicates how the specification is changed relative to the main specification, presented in column (1), where the 2012 survey flood exposure measure is recentered around its predicted value. Columns (2)-(5) show effects of the binary 2012 survey flood measure with no recentering and varying controls and sample restrictions. Columns (2)-(4) keep the main sample restrictions: communities within the common support of average flood risk stratified by the number of years of survey-reported flooding from 2007-2011, and within the common support of the predicted 2012 flood exposure measure by actual flood exposure. Column (2) adds predicted 2012 survey flood exposure-by-round controls. Column (3) adds average 100-year flood depth within 5 km-by-round and community 2012 rainfall deviation-by-round controls. Column (4) includes no additional controls. Column (5) also includes no additional controls and expands the sample to include all communities (EAs), with no exclusion based on common support of average flood risk or predicted 2012 flooding. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to alternative specifications

	Main	Add unbalanced controls	Add rain controls	Round FE
	(1)	(2)	(3)	(4)
Any food insecurity in last 12 months	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.02)	0.02 (0.03)
Daily HH consumption per capita (USD)	-0.65*** (0.22)	-1.00*** (0.32)	-0.63*** (0.21)	-0.56** (0.24)
Total value of HH farm production (USD 100s)	-3.13* (1.74)	-3.25* (1.97)	-3.21* (1.74)	-3.38* (1.87)
Total non-farm HH enterprise earnings (USD 100s)	-4.00 (3.01)	-4.41 (3.52)	-4.01 (2.99)	-3.59 (3.10)
Total wage employment earnings (USD 100s)	4.69 (5.39)	4.56 (8.38)	4.77 (5.37)	3.49 (5.48)
HH farm share of total HH earnings and prod value	-0.00 (0.01)	-0.03 (0.02)	-0.00 (0.01)	-0.01 (0.02)
NFE share of total HH earnings and prod value	0.01 (0.02)	0.02 (0.02)	0.01 (0.02)	0.03 (0.02)
Wage share of total HH earnings and prod value	-0.02 (0.01)	-0.01 (0.01)	-0.02 (0.01)	-0.02* (0.01)
HH earnings Herfindahl-Hirschman Index	0.00 (0.01)	-0.00 (0.01)	0.00 (0.01)	0.01 (0.01)
Count of HH members wkg. in HH farm	0.08 (0.09)	0.02 (0.11)	0.08 (0.09)	0.09 (0.09)
Count of HH members wkg. in HH non-farm enterprise	0.06 (0.06)	0.00 (0.07)	0.06 (0.06)	0.07 (0.05)
Count of HH members wkg. in wage employment	0.14*** (0.05)	0.17** (0.07)	0.14*** (0.05)	0.16** (0.07)
Any member left HH for work during/since last round	-0.01 (0.01)	0.00 (0.01)	-0.01 (0.01)	-0.01 (0.02)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. The columns show average effects of flood exposure across all post-flood survey rounds. Flood exposure is measured using the main survey definition of any 2012 flood report in any household survey module or in the community survey, and is recentered around predicted survey-reported 2012 community flood exposure. Each column indicates how the specification is changed relative to the main specification, presented in column (1), which includes household and state-by-round fixed effects but no controls. In column (3) we add controls for baseline characteristics that differ significantly by recentered flood status in [Table A4](#), interacted with survey round. In column (3) we instead add controls for the deviation in annual total precipitation relative to the historical mean at the community location. Columns (4) replace the state-by-round FE with round FE. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to alternative samples

	Main	All EAs	W/in 50 km of any 2012 flood report	Main, drop 20km donut	Main, drop LGA donut
	(1)	(2)	(3)	(4)	(5)
Any food insecurity in last 12 months	-0.02 (0.02)	-0.01 (0.02)	-0.01 (0.02)	-0.03 (0.02)	-0.03 (0.02)
Daily HH consumption per capita (USD)	-0.65*** (0.22)	-0.63*** (0.21)	-0.61*** (0.21)	-0.61*** (0.22)	-0.49** (0.22)
Total value of HH farm production (USD 100s)	-3.13* (1.74)	-3.45** (1.76)	-3.35* (1.77)	-4.03** (1.88)	-5.22*** (1.60)
Total non-farm HH enterprise earnings (USD 100s)	-4.00 (3.01)	-4.10 (2.88)	-3.91 (2.92)	-2.87 (3.15)	-3.74 (2.94)
Total wage employment earnings (USD 100s)	4.69 (5.39)	4.51 (5.40)	4.67 (5.47)	5.48 (4.94)	5.54 (5.37)
HH farm share of total HH earnings and prod value	-0.00 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.02)	-0.00 (0.01)
NFE share of total HH earnings and prod value	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)
Wage share of total HH earnings and prod value	-0.02 (0.01)	-0.02* (0.01)	-0.02* (0.01)	-0.02 (0.01)	-0.02 (0.01)
HH earnings Herfindahl-Hirschman Index	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.01 (0.01)
Count of HH members wkg. in HH farm	0.08 (0.09)	0.10 (0.08)	0.09 (0.08)	0.06 (0.09)	0.11 (0.09)
Count of HH members wkg. in HH non-farm enterprise	0.06 (0.06)	0.03 (0.06)	0.03 (0.06)	0.09 (0.06)	0.08 (0.06)
Count of HH members wkg. in wage employment	0.14*** (0.05)	0.15*** (0.05)	0.16*** (0.05)	0.15*** (0.05)	0.11** (0.06)
Any member left HH for work during/since last round	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.02 (0.01)	-0.01 (0.01)
Observations	11691	12328	12039	10770	10088

Note: This table repeats [Table A8](#) but varies the sample instead of the specifications. Each column indicates how the sample is changed relative to the main specification, presented in column (1), which restricts the sample to communities within the common support of average 100-year flood depth within 5 km after stratifying by the number of years with any survey flood report from 2007-2011. In column (2) the community (EA) sample restrictions are removed. Column (3) replaces the main sample restriction with dropping any community more than 50 km from any community where floods were reported in 2012. Columns (4) and (5) add sample restrictions dropping communities which may be most affected by flood exposure spillovers: those less than 20 km from any community where a flood was reported in 2012, and those in the same LGA as a flooded sample community. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Sensitivity of estimated impacts of survey-reported 2012 community flood exposure to different standard error clustering

	(1) Daily HH cons./cap.	(2) HH farm prod. val.	(3) HH crop prod. val.	(4) HH NFE earnings	(5) HH wage earnings	(6) Ct. wage wkrs.
Flood Exposure X Post	-0.648*** (0.217)	-3.131* (1.743)	-4.497*** (1.499)	-4.001 (3.012)	4.693 (5.387)	0.144*** (0.054)
SE: LGA	0.218	1.705	1.435	2.937	5.158	0.055
SE: Comm-Round	0.212	1.421	1.027	2.871	-	-
SE: Conley (50km)	0.219	1.366	1.186	2.537	3.841	0.040
SE: Conley (100km)	0.209	1.352	1.208	2.517	3.145	0.040
Wild p-val: State	0.038	0.090	0.006	0.195	0.436	0.021
Observations	11691	11691	11691	11691	11691	11691

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. Each regression follows the main specification, but we show how the standard error changes under different clustering approaches. The main regressions cluster SEs at the level of the community of residence in 2012, at which the flood exposure treatment is assigned. We then show SEs clustered at the LGA level, two-way clustered SEs at the community by round level, Conley SEs allowing for spatial correlation within 50 and 100 km, and p -values from a wild bootstrap of state-level clustered SEs. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Differences in coping strategies by flood exposure among households reporting shocks

	Non-flooded		Flood shock
	N	Mean (SD)	(SE)
Sale of livestock	1888	0.22 [0.42]	0.15*** (0.04)
Sale of land/property	1888	0.16 [0.36]	0.09** (0.03)
Sent kids away	1888	0.05 [0.22]	0.03 (0.02)
Withdrew kids from school	1888	0.04 [0.19]	0.01 (0.02)
Increased enterprise activity	1888	0.05 [0.22]	0.04* (0.02)
Assistance from friends/family	1888	0.28 [0.45]	0.05 (0.04)
Borrowed from friends/family	1888	0.23 [0.42]	0.03 (0.04)
HH member migrated for work	1888	0.03 [0.17]	0.02 (0.02)
Took loan or credit	1888	0.18 [0.39]	-0.00 (0.03)
Reduced consumption	1888	0.22 [0.41]	0.06* (0.03)
Received other assistance	1888	0.01 [0.12]	0.03** (0.02)

Note: This table shows the results from separate regressions of indicators that households reporting any shock in the 2013 post-harvest survey declare different coping strategies on an indicator for whether they declare any flood shock, controlling for whether they report other types of shocks. All regressions include community fixed effects. Standard errors are clustered by community. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Impacts of survey-reported 2012 community flood exposure on household consumption and assets

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Daily HH consumption per adult equiv (USD)	11676	6.52 [4.91]	-0.61*** (0.23)	-0.62* (0.36)	-0.60** (0.25)	-0.63* (0.34)
Daily HH consumption per capita (USD)	11691	5.64 [4.40]	-0.65*** (0.22)	-0.83*** (0.30)	-0.47* (0.25)	-0.46 (0.30)
Daily HH food cons. per capita (USD)	11691	4.35 [4.43]	-0.64*** (0.18)	-0.69*** (0.21)	-0.58*** (0.22)	-0.64*** (0.24)
Daily HH non-food cons. per capita (USD)	11691	1.29 [2.79]	-0.01 (0.18)	-0.14 (0.27)	0.11 (0.20)	0.19 (0.28)
HH under 1.90 USD PPP per capita daily poverty line	11691	0.32 [0.47]	0.05** (0.02)	0.05** (0.02)	0.04 (0.03)	0.08** (0.03)
Value of HH assets (USD)	11691	2356.56 [5520.11]	-86.71 (152.13)	-94.69 (173.28)	-94.96 (176.48)	5.45 (367.08)
Livestock holdings (TLU)	11691	2.72 [42.95]	1.57* (0.87)	1.41 (0.94)	1.74** (0.86)	1.66** (0.78)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A13: Impacts of survey-reported 2012 community flood exposure on household earnings and production value by activity

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Total HH earnings and production value (USD 100s)	11691	89.42 [205.00]	-0.99 (6.50)	3.45 (7.62)	-5.16 (6.70)	-5.43 (11.94)
Total value of HH farm production (USD 100s)	11691	18.12 [40.83]	-3.13* (1.74)	-4.02** (1.88)	-2.17 (2.00)	-2.92 (2.72)
Total crop production value (USD 100s)	11691	15.18 [32.69]	-4.50*** (1.50)	-4.68*** (1.63)	-4.40** (1.81)	-3.91 (2.55)
Total livestock production value (USD 100s)	11691	2.93 [23.19]	1.37* (0.82)	0.66 (0.82)	2.23** (0.87)	0.98 (1.49)
Total earnings from non-farm activities (USD 100s)	11691	71.31 [204.28]	2.14 (6.38)	7.48 (7.41)	-2.99 (6.63)	-2.51 (12.06)
Total non-farm HH enterprise earnings (USD 100s)	11691	26.83 [77.46]	-4.00 (3.01)	-1.81 (3.26)	-6.97** (3.44)	-1.32 (5.17)
Total wage employment earnings (USD 100s)	11691	40.63 [188.67]	4.69 (5.39)	7.48 (6.40)	2.25 (5.45)	1.02 (10.08)
Total HH earnings from other activities (USD 100s)	11691	3.85 [22.79]	1.45* (0.84)	1.81** (0.89)	1.74 (1.07)	-2.21 (1.79)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A14: Impacts of survey-reported 2012 community flood exposure on household migration and roster changes

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Any member left HH during/since last round	11691	0.09 [0.28]	-0.01 (0.02)	-0.02 (0.02)	0.02 (0.02)	-0.08** (0.03)
Any member left HH for work during/since last round	11691	0.02 [0.13]	-0.01 (0.01)	-0.01 (0.01)	-0.03** (0.02)	0.06* (0.03)
Any member left HH for family during/since last round	11691	0.03 [0.16]	-0.03** (0.02)	-0.04** (0.02)	-0.03 (0.02)	-0.05 (0.04)
Any adult member joined HH during/since last round	11691	0.08 [0.27]	0.02 (0.01)	0.02 (0.01)	0.02* (0.01)	-0.00 (0.02)
Count of HH members	11691	5.36 [2.83]	0.03 (0.06)	0.02 (0.05)	0.03 (0.07)	0.11 (0.20)
Count of HH members ages 0-4	11691	0.80 [1.04]	-0.04 (0.04)	-0.04 (0.04)	-0.04 (0.05)	-0.06 (0.09)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. See [Appendix C](#) for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Impacts of survey-reported 2012 community flood exposure on wages and market food prices

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
HH avg. PP daily wage earned by wage wkrs (USD)	1964	27.82 [30.64]	-6.75** (2.91)	-4.94 (3.48)	-8.45** (3.43)	-12.06* (6.46)
HH avg. PH daily wage earned by wage wkrs (USD)	2072	32.78 [43.35]	6.24 (4.19)	9.57* (5.42)	3.85 (4.99)	-3.48 (5.23)
HH avg. daily wage paid to hired ag labor (USD)	4417	12.92 [16.89]	2.24 (1.57)	1.87 (1.78)	2.58* (1.50)	2.45 (1.61)
Comm. avg. daily wage for men's ag labor (USD)	9224	17.41 [9.20]	-0.48 (0.92)	-0.75 (1.20)	-0.31 (0.94)	0.35 (1.74)
Comm. price shelled maize (USD/kg z-score)	8259	-0.10 [0.63]	0.10 (0.14)	0.25 (0.17)	-0.08 (0.12)	
Comm. price local rice (USD/kg z-score)	8530	-0.10 [0.54]	-0.06 (0.10)	0.15 (0.14)	-0.26** (0.13)	
Comm. price bread (USD/kg z-score)	9933	-0.12 [0.65]	-0.32*** (0.11)	-0.29** (0.13)	-0.35*** (0.13)	
Comm. price yam roots (USD/kg z-score)	9016	0.00 [0.95]	-0.07 (0.12)	0.01 (0.13)	-0.14 (0.14)	
Comm. price palm oil (USD/litre z-score)	9924	0.04 [1.15]	0.00 (0.10)	-0.08 (0.12)	0.09 (0.12)	
Comm. price groundnut oil (USD/litre z-score)	9475	0.07 [1.35]	0.12 (0.10)	0.13 (0.11)	0.11 (0.12)	
Comm. price banana (USD/kg z-score)	7625	0.03 [0.96]	0.11 (0.16)	-0.07 (0.16)	0.30 (0.20)	
Comm. price chicken (USD/kg z-score)	8313	0.06 [1.02]	0.05 (0.11)	-0.16 (0.11)	0.32** (0.16)	
Comm. price sugar (USD/kg z-score)	8281	-0.06 [0.85]	0.11 (0.13)	0.09 (0.15)	0.12 (0.13)	

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. Household wages received or paid are only reported by households active in wage labor markets. Market food prices were not collected in the same way in 2018-19, so estimates for these outcomes exclude that round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A16: Impacts of survey-reported 2012 community flood exposure on average household physical yield and sales prices by crop

	N	Control Mean (SD)	2012 survey recentered × Post (SE)	2012 survey recentered × 2012-13 (SE)	2012 survey recentered × 2015-16 (SE)	2012 survey recentered × 2018-19 (SE)
Yield by area planted of cereals (kgs/ha)	4151	2181.91 [4310.20]	-1082.85** (512.46)	-1454.12** (592.10)	-685.44 (578.58)	-583.84 (823.92)
Yield by area planted of maize (kgs/ha)	3048	1419.84 [3525.54]	-613.29 (411.88)	-1343.98** (528.61)	-24.08 (488.58)	-792.79* (475.45)
Yield by area planted of sorghum (kgs/ha)	2869	1685.30 [3246.21]	-236.93 (422.54)	-287.28 (487.21)	-178.02 (558.96)	-239.79 (615.93)
Yield by area planted of millet (kgs/ha)	1792	1170.63 [1605.19]	-828.54 (557.41)	-917.29 (678.87)	-795.51 (716.08)	-105.53 (602.26)
Yield by area planted of rice (kgs/ha)	475	2714.51 [2601.40]	-1583.35* (911.22)	-1131.83 (1313.15)	-1583.95 (971.16)	-5600.71*** (1979.35)
Yield by area planted of cowpea (kgs/ha)	2000	593.18 [1196.41]	-483.13** (211.35)	-478.10* (249.49)	-502.53* (295.09)	-296.93 (391.35)
Yield by area planted of groundnut (kgs/ha)	858	1268.76 [2587.19]	-1005.89** (492.16)	-923.31 (599.89)	-1187.54 (926.94)	-577.87 (1604.91)
Yield by area planted of yam (kgs/ha)	2421	8899.53 [18059.75]	-1692.69 (2514.69)	-4471.32 (3703.42)	704.08 (3096.73)	4121.74 (3244.46)
Yield by area planted of cassava (kgs/ha)	3254	1800.41 [4145.43]	764.96 (832.23)	1094.43 (1159.13)	219.97 (951.95)	1819.89 (1228.69)
Average cereal sales price (USD/kg)	4141	0.55 [1.12]	-0.10 (0.08)	-0.08 (0.07)	-0.12 (0.12)	-0.06 (0.20)
Average maize sales price (USD/kg)	2544	0.31 [2.51]	0.11 (0.14)	0.10 (0.16)	0.11 (0.14)	0.23 (0.17)
Average sorghum sales price (USD/kg)	2814	0.99 [0.93]	0.02 (0.09)	0.06 (0.06)	-0.05 (0.15)	0.30** (0.15)
Average millet sales price (USD/kg)	1766	0.14 [0.55]	-0.19 (0.13)	-0.02 (0.07)	-0.37 (0.23)	-0.18 (0.16)
Average rice sales price (USD/kg)	412	0.96 [0.54]	-0.21** (0.09)	-0.22 (0.20)	-0.21 (0.20)	-0.17 (0.19)
Average cowpea sales price (USD/kg)	1902	1.72 [1.01]	0.04 (0.28)	0.00 (0.34)	0.06 (0.21)	0.49 (0.35)
Average groundnut sales price (USD/kg)	801	0.02 [0.22]	0.07 (0.07)	0.07 (0.10)	0.10 (0.09)	-0.09 (0.12)
Average yam sales price (USD/kg)	2277	0.66 [0.50]	-0.03 (0.02)	-0.00 (0.05)	-0.03 (0.06)	-0.21 (0.17)
Average cassava sales price (USD/kg)	976	0.68 [1.20]	0.33 (0.27)	0.25 (0.29)	0.42 (0.31)	0.46 (0.29)

Note: This table shows the results from separate regressions of household outcomes on recentered 2012 community flood exposure interacted with being observed after 2012. In this table, flood exposure is defined based on whether any household reports being affected by flooding or the community survey reports floods that negatively affected community members in 2012 in the GHSP. Each row represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. We show results for the eight most commonly-cultivated crops in Nigeria across the full sample. Sample sizes vary based on the number of households cultivating each crop. Sample sizes for prices are smaller than for yields because of households not harvesting any of the crop they planted. Control means are for communities with no 2012 survey-based flood exposure in the 2010-11 survey round. The first column of results shows average effects across all post-flood survey rounds. The next three columns show dynamic effects in each post-flood round. All regressions include household and state-by-round fixed effects. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A17: Heterogeneity in impacts of survey-reported 2012 community flood exposure by baseline household and community characteristics

		HH agriculture		Wage work		HH NFE	
	(1) Daily HH cons./cap.	(2) Value prod.	(3) Count mems.	(4) Total earn.	(5) Count mems.	(6) Total earn.	(7) Count mems.
Flood $\times$ Post $\times$ > 5 baseline HH members	0.18 (0.38)	-5.29 (3.33)	-0.06 (0.16)	-17.52* (9.96)	0.22** (0.10)	8.64 (5.91)	0.08 (0.10)
Flood $\times$ Post $\times$ Above med. baseline HH assets value	-0.10 (0.37)	-1.14 (3.52)	-0.13 (0.13)	15.73 (10.87)	-0.12 (0.10)	-0.68 (6.07)	-0.12 (0.10)
Flood $\times$ Post $\times$ Rural community	-0.32 (0.49)	-1.57 (2.72)	0.18 (0.15)	11.11 (16.87)	0.31*** (0.10)	14.98* (7.72)	0.24** (0.10)
Flood $\times$ Post $\times$ Below med. dist. to major market	-0.65 (0.48)	3.86 (3.49)	-0.01 (0.19)	0.79 (10.63)	-0.04 (0.12)	-15.11** (6.47)	-0.18 (0.12)
Flood $\times$ Post $\times$ Below med. dist. to major road	0.07 (0.41)	-1.04 (3.14)	-0.22 (0.16)	23.56** (10.78)	-0.02 (0.10)	-9.77* (5.74)	-0.11 (0.11)
Flood $\times$ Post $\times$ Below med. dist. to > 20k pop. town	0.49 (0.42)	2.12 (3.22)	-0.12 (0.17)	3.74 (10.25)	-0.11 (0.10)	-11.30* (5.83)	-0.06 (0.12)
Flood $\times$ Post $\times$ Below med. dist. to river/water	-0.49 (0.46)	-6.55* (3.62)	0.21 (0.17)	-1.17 (11.29)	-0.03 (0.11)	0.14 (5.98)	-0.05 (0.12)
Flood $\times$ Post $\times$ Non-zero fluvial flood risk	-0.22 (0.58)	-1.83 (3.30)	-0.21 (0.18)	-2.35 (13.88)	0.02 (0.11)	-2.28 (6.51)	-0.24* (0.12)
Flood $\times$ Post $\times$ Comm. flooded in last 5 years	0.43 (0.58)	-2.82 (4.03)	-0.41 (0.27)	0.65 (9.97)	-0.04 (0.16)	12.27** (5.31)	-0.24 (0.15)

Note: This table presents tests of heterogeneity in average effects of 2012 community flood exposure on household outcomes in subsequent survey rounds by baseline household and community characteristics. Each column represents an outcome. Monetary values are in 2016 PPP USD. See [Appendix C](#) for details on variable construction. The results for each column in each row are from separate triple-differences regressions fully interacting the indicated baseline household or community characteristic with the Post and Flood dummies and state-by-round fixed effects. We show only the estimated coefficient for the interaction term for simplicity, reflecting the estimated difference in effects for households/communities with a given characteristic. Standard errors are clustered at the level of the community of residence in 2012. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A18: Balance in baseline characteristics by baseline livelihood activities

	Non-ag Mean (SD)	Ag HH difference (SE)	Ag HH, Non-NFE Mean (SD)	NFE HH difference (SE)	Ag HH, Non-wage Mean (SD)	Wage HH difference (SE)
<i>Community characteristics</i>						
Rural	0.26 (0.44)	0.50*** (0.03)	0.89 (0.32)	-0.04*** (0.02)	0.87 (0.33)	-0.09*** (0.02)
HH distance to nearest major road (km)	7.76 (14.31)	5.05*** (1.25)	18.39 (19.80)	-2.66*** (0.92)	18.29 (20.07)	-2.93*** (0.87)
HH distance to nearest market (km)	55.58 (48.72)	10.90*** (2.88)	70.47 (40.07)	-3.02* (1.80)	72.30 (40.57)	-2.97* (1.55)
HH distance to nearest pop. center w/ ≥20k pop. (km)	9.57 (12.78)	9.48*** (1.24)	24.85 (21.13)	-2.63*** (0.87)	24.82 (21.33)	-3.02*** (0.87)
Percent ag land w/in approx 1km	14.61 (19.88)	10.33*** (1.74)	29.52 (24.95)	1.37 (0.89)	35.02 (28.55)	-0.74 (0.95)
Slope (percent)	3.11 (2.61)	0.28 (0.21)	3.21 (2.35)	-0.01 (0.10)	3.24 (2.78)	-0.14 (0.12)
Elevation (m)	216.69 (200.01)	9.14 (8.35)	274.27 (199.68)	4.14 (5.15)	305.27 (199.81)	12.26* (6.32)
Annual Precipitation (mm)	1608.39 (627.51)	-38.80*** (14.02)	1421.39 (612.53)	9.91 (8.57)	1334.49 (593.59)	-0.75 (7.24)
Distance to nearest water area (km)	32.44 (31.40)	7.57*** (1.94)	33.41 (28.28)	0.00 (1.25)	33.61 (29.39)	1.24 (1.02)
Avg 100-year flood depth within 5 km	0.38 (0.65)	-0.10** (0.04)	0.17 (0.42)	-0.02 (0.02)	0.17 (0.39)	-0.01 (0.01)
Any comm. HH flood shock rept. from 2007-2011	0.09 (0.29)	0.10*** (0.03)	0.24 (0.43)	0.03 (0.02)	0.30 (0.46)	0.01 (0.02)
<i>Household characteristics</i>						
Female-headed HH	0.20 (0.40)	-0.06*** (0.02)	0.17 (0.38)	-0.05*** (0.01)	0.14 (0.34)	-0.06*** (0.01)
Count of HH members	4.62 (2.49)	0.89*** (0.13)	5.21 (3.08)	0.94*** (0.13)	5.68 (3.01)	0.95*** (0.12)
Household asset index	0.66 (1.22)	-0.65*** (0.06)	-0.29 (0.95)	0.11** (0.04)	-0.34 (0.67)	0.36*** (0.04)
Any food insecurity in last 12 months	0.20 (0.40)	-0.01 (0.02)	0.21 (0.41)	-0.03 (0.02)	0.18 (0.38)	-0.01 (0.01)
Daily HH consumption per capita (USD)	7.97 (6.24)	-2.63*** (0.26)	4.90 (4.42)	-0.34* (0.18)	4.42 (3.48)	0.28** (0.14)
<i>Household activities</i>						
Any HH non-farm enterprise activity	0.81 (0.39)	-0.02 (0.02)	0.00 (0.00)	1.00*** (0.00)	0.76 (0.43)	0.04*** (0.02)
Any wage employment activity	0.53 (0.50)	-0.16*** (0.02)	0.33 (0.47)	0.06*** (0.02)	0.00 (0.00)	1.00*** (0.00)
Total value of HH farm production (USD 100s)	0.00 (0.00)	26.21*** (1.94)	23.27 (47.52)	2.69 (2.05)	25.39 (38.95)	-0.74 (1.61)
Total non-farm HH enterprise earnings (USD 100s)	52.31 (121.67)	-30.09*** (4.86)	0.00 (0.00)	22.09*** (1.61)	17.77 (58.13)	3.19 (2.45)
Total wage employment earnings (USD 100s)	79.26 (234.13)	-55.43*** (10.52)	16.92 (54.75)	3.93 (3.71)	0.00 (0.00)	50.77*** (5.40)
Total area planted (ha)	0.00 (0.00)	1.97*** (0.15)	1.94 (3.42)	0.17 (0.15)	2.39 (3.83)	0.03 (0.15)

Note: This table shows the baseline (2010-11) control mean and difference by baseline livelihood activities for selected community and household characteristics from a series of separate regressions for the main analysis sample. All regressions include state fixed effects, and standard errors are clustered at the community level. We show three sets of balance tests. The first two columns show differences by baseline engagement in agriculture across all sample columns. The next two columns show differences by baseline engagement in NFE among households engaged in agriculture at baseline. The last two columns show differences by baseline engagement in wage work among households engaged in agriculture at baseline. In all three sets of columns, the first column shows the mean for the reference group and the second column shows the coefficient on the particular baseline activity from a regression of that activity on the outcome at in 2010-11. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C Survey Data

The main data source for the analysis is Nigeria’s General Household Survey Panel (GHSP). The GHSP, part of the World Bank’s Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA), is a nationally-representative panel survey including 5,000 households conducted by the Nigeria National Bureau of Statistics. Before the first survey round, 500 enumeration areas (communities) were randomly sampled after stratifying by state. Ten households were then randomly sampled in each enumeration area from a census of households.

Four survey rounds were conducted between 2010 and 2019. Households are tracked over time, including if they move to a new location, but individual household members are not tracked. In 2018-19 the panel sample was partially refreshed, with 1,425 households from the original panel retained (clustered in randomly selected panel communities) and 3,551 new panel households added to the sample. The GHSP data are publicly available from the World Bank’s Microdata Catalog.

Each survey round includes both a post-planting and a post-harvest survey. Post-planting surveys took place in September-October 2010, September-October 2012, August-September 2015, and July-August 2018. Post-harvest surveys took place in February-April of 2011, 2013, and 2016 and in January-February 2019. Some information, such as individual labor supply, is recorded in each survey while other modules are only included in either the post-planting or post-harvest surveys.

Basic cleaning decisions include replacing missing values with 0 where appropriate (i.e., for income in an activity the household was not engaged in) and replacing impossible values (such as more than 24 hours per day) with missing values. We winsorize continuous variables by replacing values above the 99th percentile with the value at the 99th percentile. All variables representing monetary values are converted to 2016 PPP USD.

For certain variables – notably the measures of total household earnings by activity – we modify code developed by the Evans School Policy Analysis & Research Group (EPAR) to construct variables in consistent ways accounting for differences in the survey instruments across rounds. The code is available on GitHub: <https://github.com/EvansSchoolPolicyAnalysisAndResearch/LSMS-Agricultural-Indicators-Code>.

Construction of outcome variables

We report impacts on a wide variety of outcomes. Daily household consumption outcomes are calculated on the basis of reported consumption across different goods and services, and are provided directly in the publicly-available GHSP data. Daily consumption is divided by either the count of household members or the count of adult equivalents. Poverty is calculated based on per capita daily consumption. The value of household assets is the sum of declared values across all assets owned. Livestock are measured in tropical livestock units (TLU) in order to aggregate across species.

Changes in household composition are measured in reference to the previous survey round, as

the household roster is updated in each survey and respondents can indicate which members have left and their reason for leaving and if any new members have joined. Counts of household members working in different activities are based on the household labor and agricultural labor modules in the post-planting and post-harvest surveys where respondents indicate which members are engaged in different activities. We code each member as engaged in a given activity for a survey round if they were engaged in it during either the post-planting or post-harvest round, then take the sum of members in each activity for the household. Hours of work over the last 7 days by household members are aggregated from the same survey modules.

Section 3.1 describes how the earnings and value of production variables are constructed. Farm production includes crop and livestock activities. Non-farm earnings include household non-farm enterprise, wage employment, and ‘other activities’ including income from pensions, remittances, investments, transfers, etc. Households not active in a given activity are assigned an income of zero for that activity. We drop households with zero total earnings and production value in 2010-11 from the analysis as these likely represent missing data rather than true zeros. Households with zero total earnings and production value are more prevalent in 2010-11 (11.1% of the sample) than in subsequent rounds (3.8%) and do report being engaged in farm production (43%), NFE (51%), and wage employment (23%), indicating challenges with collecting earnings or production data in the first implementation of the survey. Results are qualitatively similar but with slightly smaller coefficient magnitudes if these households are included. The household earnings Herfindahl-Hirschman-Index (HHI) is the sum of squared shares of earnings (without subtracting expenses) from farm production, NFE, wage work, and other sources. Values closer to one indicate a greater concentration of income sources.

All crop production outcomes refer to the most recent growing season and are aggregated across plots. Outcomes for households not active in crop production in a given round are assigned values of 0, except for the crop area HHI and proportion of crop value sold which are undefined for households with 0 area planted and the crop yield and price values which are undefined for households not cultivating particular crops. The number of crops cultivated is across all household plots. The crop area HHI is the sum of squared shares of total planted area allocated to different crops. Values closer to 1 indicate less diversification across crops. Total hired/exchange and family labor days and family labor hours are summed across all crop production activities reported in the post-planting and post-harvest surveys. Crop production costs are the sum of expenditures on purchased inputs such as seeds, fertilizer, labor, land, etc. Crop sales are the sum across crops of quantity sold times sales price. Crop production value is the sum across crops of crop sales plus the value of unsold crops, measured as quantity harvested but not sold times either the median household sale price, the median sales price for that crop in the community if the household did not sell but at least three community households did, or the median price at the lowest administrative level with at least three sales observations. The proportion of value of crop production sold is the ratio of the two above variables.

Physical crop yields are calculated as the total quantity harvested divided by the area planted.

Crop sales prices are defined in the calculation of crop production value. For cereals, we calculated mean yields by taking total cereal quantity harvested over total cereal area planted and calculated quantity-weighted mean prices across different cereals produced. We calculate a household crop yield index and crop price index across the eight most commonly cultivated crops in the sample: maize, rice, sorghum, millet, cowpea, groundnut, yam, and cassava. We first normalize yield and sales price within survey round for each crop, and then calculate area planted-weighted means of normalized yields and prices across crops produced. We then normalize these indices within survey round.

Differences across survey flood measures

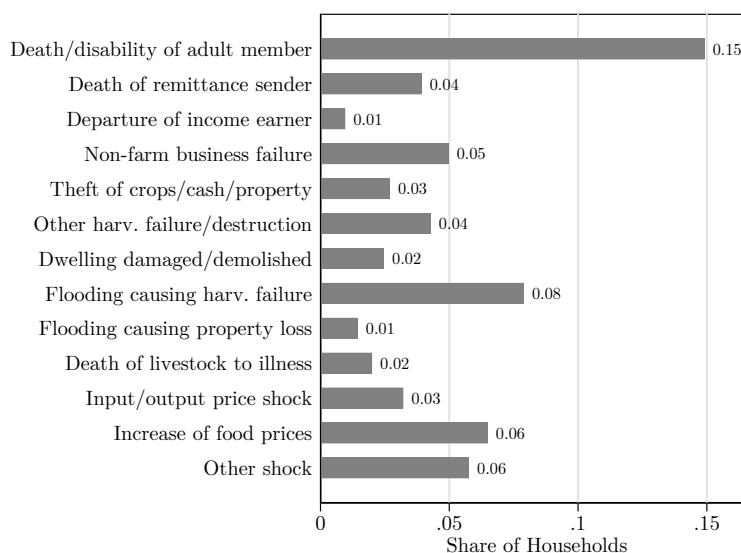
Several survey questions ask about flood exposure, each of which captures specific ways in which floods can affect households. In the post-planting agricultural survey, households can list flooding as the main cause of loss of stored crops since the beginning of the new year for each cultivated crop. Such losses are reported by 13% of households for 2012 but flood-related losses are only reported by 1% In the post-harvest agricultural survey, households can list flooding as a reason for not harvesting a particular crop. Failure to harvest is reported by 24% of households, but floods are only listed as a reason by 3%. In both the post-planting and post-harvest household surveys, households can list flooding as a cause of household food insecurity in the past 12 months. Food insecurity is reported by 24% of households at post-planting and by 20% at post-harvest in the 2012-13 survey, and in both periods 3% list floods as a cause of food insecurity. Finally, in the post-harvest household survey shocks module, households are asked about various shocks they have been affected by over the past 5 years, the years in which they occurred, and the consequences of the shocks. Two shocks relate to floods: flooding that caused harvest failure is reported by 7% of households in 2012 and loss of property due to flood is reported by 1%. Floods are the second most commonly reported shock for 2012 (Figure A7).

We can use the fact that household shocks are reported for multiple years to show how the prevalence of flood shocks was much greater in 2012 than in any other year from 2007-2018 (Figure A8). The fact that the share of households reporting flood shocks is much lower than the share of communities where such shocks are reported highlights how not every household is affected in communities where flooding occurs.

Across all flood questions, 12.4% of households report being affected by floods in some way in 2012. ?? visualizes the count of households reporting floods across communities, further demonstrating how in most communities where floods are reported, only a subset of the ten sample households report being affected. This result is supported by the community questionnaire, which asks informants to report important events that made people worse off in the community, including floods, and the year in which they occurred. Floods are reported in 20% of communities in 2012, and on average these are reported to have affected 48% of community households (the median is 50%).

The main survey flood definition is based on the union of reports of floods from all household

Figure A7: Prevalence of shocks reported by households over the 12 months before the 2013 post-harvest survey round



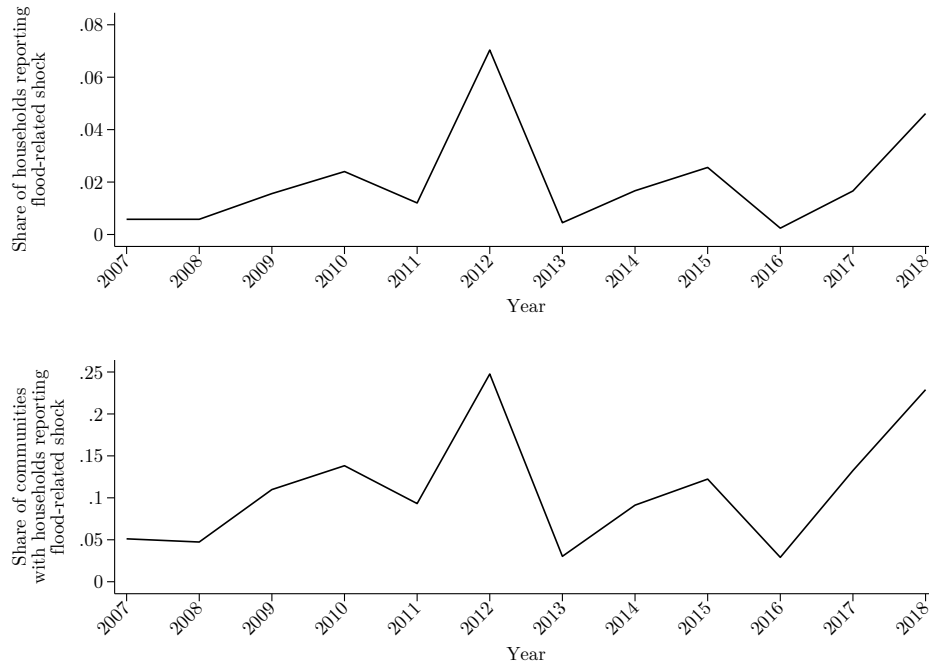
Note: Data are from the 2013 post-harvest household survey shocks module.

survey questions and from the community survey. [Figure A10](#) shows a heatmap of correlations between these measures. We aggregate the agricultural survey questions as ‘hh_harv_loss’ and the food insecurity questions as ‘hh_food_insec’, in both cases coded as 1 if the household reports being affected by a flood in either the post-planting or post-harvest wave. We compare these against the two flood questions from the household shock module (‘hh_harv_shock’ and ‘hh_property_shock’) and against reports of floods that negatively affected community members by informants in the community survey (‘comm_flood’). While the two harvest-related measures are strongly correlated, correlations between the other measures are weaker. This emphasizes that each measure is capturing different ways in which floods may affect households, and that they primarily affect different households rather than the same household being affected in multiple ways. The figure also indicates that household flooding reported in the shocks module – a common way survey-based flood measures are constructed – may miss some floods that did not cause the specific types of losses or damages that are asked about.

How do these household flood reports compare to what is reported in the GHSP community surveys? Twenty percent of households reside in communities where flooding was reported in 2012 in the community survey. The survey and community measures are correlated: in communities with a community survey flood report, 25% of households report a flood, compared to 7% in communities with no community flood report. But this mismatch leads to 41% of households that report being affected by floods in 2012 residing in communities with no community survey flood report.

Households reporting floods not captured in the community survey may be driven by lack of complete information by the group of community survey informants, or by error in the household

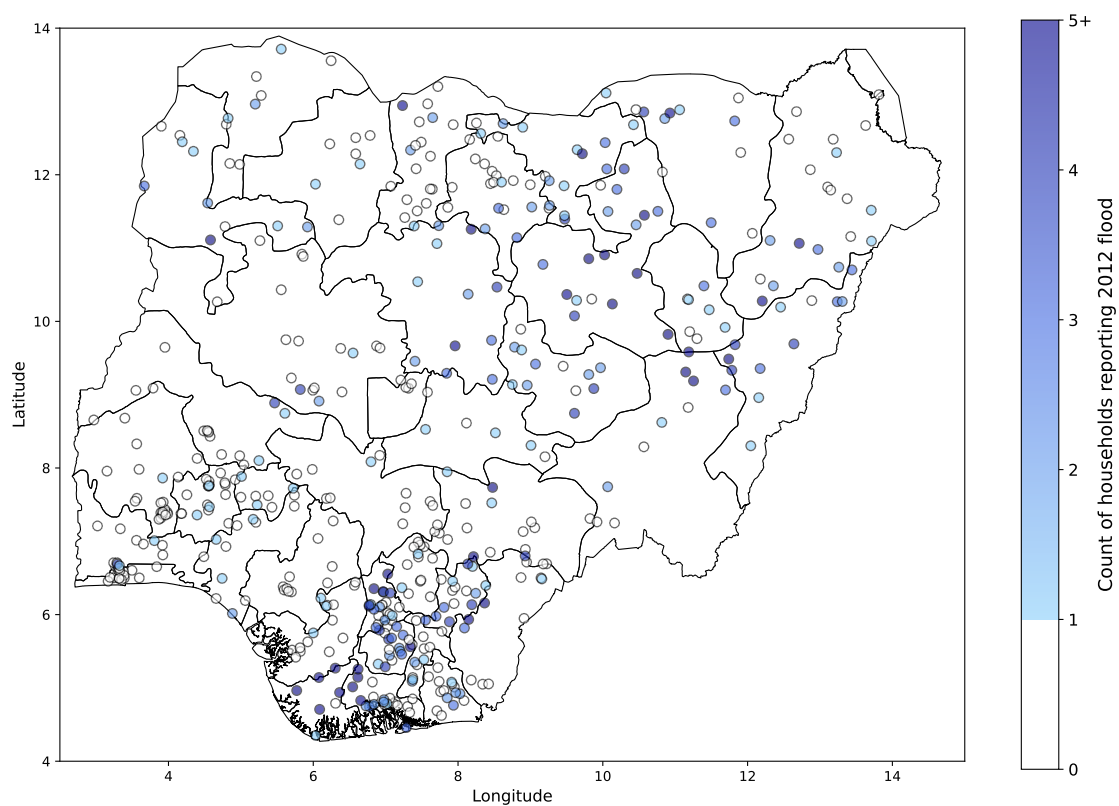
Figure A8: Survey flood shock reports by year



Note: Annual flood reports are from the Nigeria GHSP. In each survey round, households are asked to recall whether they experienced specific economic shocks over the last few years, including flooding that caused harvest failure and loss of property due to flood. These data were collected in the post-harvest survey waves in 2009, 2013, 2016, and 2019. The top panel shows the share of households reporting any flood shock by year. The bottom panel shows the share of communities with any household flood shock report by year. This is a more restrictive measure of flood exposure than the main survey-based measure we use in this paper which incorporates reports of floods in the survey's food security and crop production modules and in the community informant survey, but there is no recall element for the other household survey questions.

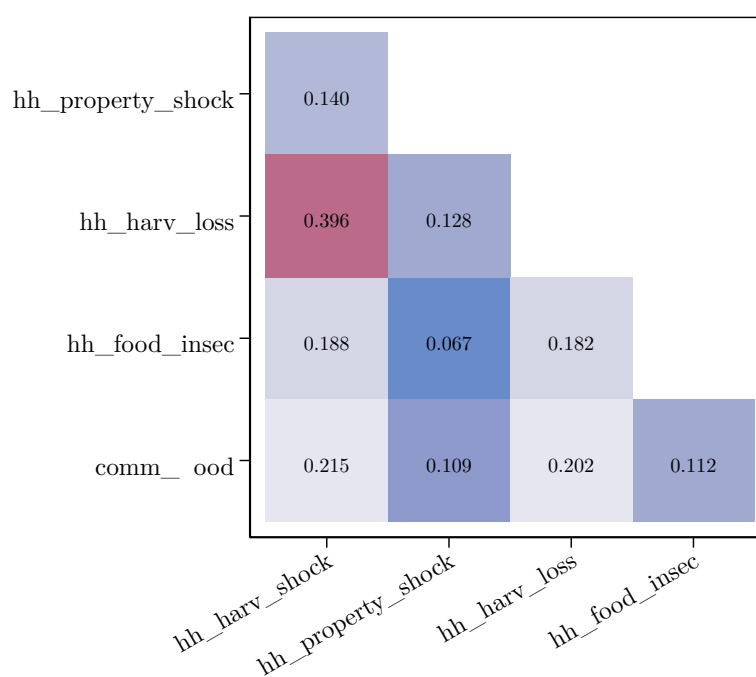
surveys. Even in communities with at least two (of ten sample households) separate household reports of flood in the shocks module, some community surveys do not report flooding. Community flood reports are more strongly correlated with household reports of harvest losses than with measures of property loss or food insecurity. This suggests differences in the types of damages that are more salient to community informants. On the other hand, there are 41 communities with a community flood report but no survey flood report. A potential reason for this is floods affecting community households other than those included in the GHSP sample, or affecting households in ways not captured in the household survey questions.

Figure A9: Counts of households reporting flood exposure in 2012 by community



Note: Nigeria GHSP household flood reports are aggregated across questions from the household shocks, food security, and crop production modules. For each community, we calculate the number of households reporting being affected by floods in 2012 in one of these modules. White circles indicating communities with no household flood reports. Ten households are sampled for each survey community.

Figure A10: Correlations between survey measures of household-level 2012 flood exposure reports



Note: This heatmap shows pairwise correlations between different measures of flood exposure in 2012 at the level of households in the Nigeria GHSP. 'hh_harv_shock' indicates a report of a flood that caused harvest failure in the household shocks module. 'hh_property_shock' indicates a report of a flood that caused property loss in the household shocks module. 'hh_harv_loss' indicates a report of loss of stored crops or failure to harvest in the post-planting and post-harvest household crop production modules. 'hh_food_insec' indicates a report of floods as a cause of household food insecurity in either the post-planting or post-harvest food security modules. 'comm_flood' indicates that informants in the community survey report that floods negatively affected community members.

D Flood detection with MODIS satellite imagery

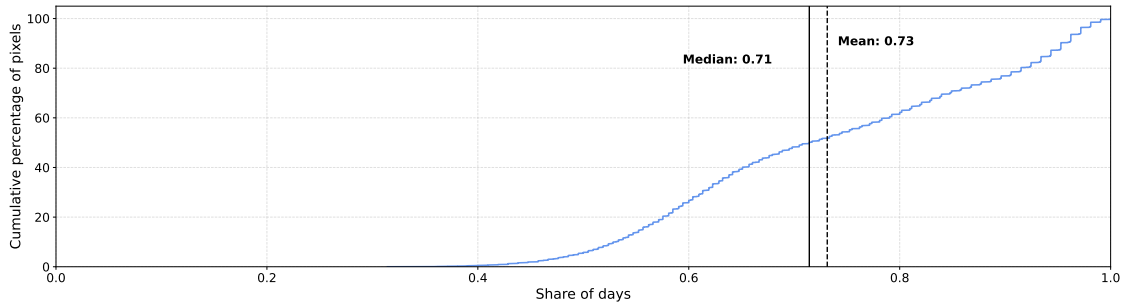
This appendix illustrates reasons why remote detection of flooding using MODIS satellite imagery, as used in this paper via the NASA Global Flood Product, may not align with floods reported in surveys. First, optical satellite imagery is constrained by cloud cover which can limit the ability to observe changes in surface water. Second, MODIS has relatively coarse spatial resolution, limiting its ability to detect narrow or localized water features. Third, the satellite-based community indicator is constructed using a binary intersection rule: a community is classified as flooded whenever the any 2012 flooding is including in the NASA Global Flood Product within a 5 km buffer. Floods detected within this buffer may not affect survey households for a variety of reasons.

D.1 Cloud cover and observability

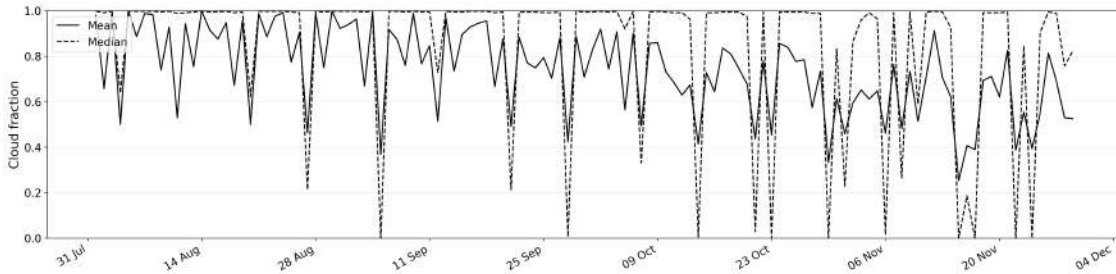
Optical satellite imagery cannot detect floods when the Earth’s surface is covered by clouds. [Figure A11](#) documents cloud conditions during the 2012 flood season in Nigeria, between August and November. Panel A shows that the median MODIS pixel (250 m) in Nigeria was cloudy on roughly 71% of days, and the mean was approximately 73%. This indicates a broad national pattern of frequent cloud cover during the months in which flooding was most likely to occur.

Figure A11: Cloud cover in MODIS satellite imagery in Nigeria, August-November 2012

(a) Distribution of share of cloudy days across all Nigeria pixels



(b) Daily mean and median cloud fraction within 5 km buffers around GHSP communities



Note: Data are from the MOD09GA collection on Google Earth Engine.

Panel B focuses on the empirical setting of the paper by plotting the daily mean and median cloud fraction within 5 km buffers around GHSP communities. Cloud cover remains high for much of the flood season, with means around 80-90% from August-September and 50-80% from October-November. The median is often close to one, implying that on most dates at least half of the

surveyed communities were almost entirely cloud-covered. Although the cloud fraction occasionally drops sharply, these clearer observation windows are brief and irregular.

Taken together, these patterns indicate that cloud cover during the period when most floods occurred in 2012 is a major constraint to remote sensing of flooding using MODIS imagery. The constraint implies that the absence of flooding in the NASA database cannot be interpreted as evidence that no flooding occurred on the ground, and likely explains a large share of the failure of the database to identify flooding in communities with survey-reported floods.

D.2 Spatial resolution and water detection

Another limitation of MODIS imagery is its 250 m resolution. Detection of surface water is based on reflection of different bandwidths of light, formalized in the Normalized Difference Water Index (NDWI). Pixels of mixed water and other surfaces will have lower NDWI values than pixels with a large share of water cover, and the water cover is therefore less likely to be identified. We illustrate how the MODIS data resolution limits its ability to detect water in comparison with Landsat 7 satellite imagery, which has a 30 m resolution but is collected less frequently.

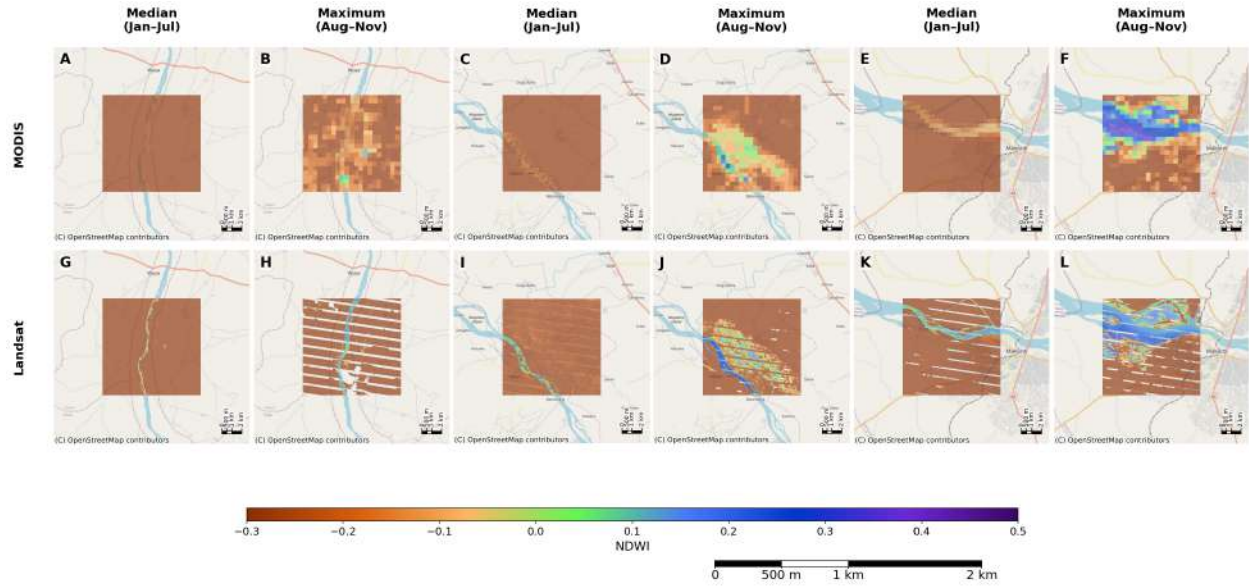
For both sources, we calculate the McFeeters NDWI (McFeeters, 1996), which is designed to enhance open surface water, rather than the Gao NDWI (Gao, 1996), which is more closely related to vegetation moisture. NDWI values above 0 or 0.1 are typically taken to likely represent surface water, with higher values indicating greater certainty. During the non-flood period (January–July 2012), we use median NDWI to represent typical surface-water conditions. During the flood period (August–November 2012), we use maximum NDWI to give the imagery the best possible chance to reveal changes in surface water.

Figure A12 shows three cases illustrating how the ability of 250 m resolution MODIS imagery to detect water depends on the spatial scale of the underlying feature, depicted using data from OpenStreetMap. In the first case, MODIS detects no meaningful variation in NDWI above the narrow river in the non-flood season (Panel A), and the signal remains unclear in the flood season (B). In the second case, with a slightly larger river, MODIS identifies a different NDWI signal during the non-flood period but it is not above typical thresholds for classifying water (C). It does identify some clear river water cells in the flood season, as well as some cells with potential flooding (D). In the third case, with a river around 500 m in width at many points even in the dry season, MODIS imagery better identifies cells with higher NDWI where the river lies during the non-flood season (E), but these NDWI values are not high enough to clearly identify water though identification is stronger during the flood season (F).

In all cases, the 30 m resolution Landsat 7 imagery (bottom row) clearly identifies water cells above the rivers in the dry non-flood season, as well as expansions and overflows of these rivers during the flood season. In all cases, the higher resolution Landsat imagery provides sharper and more precise information on river channel structure and likely flood overflows.

The fact that MODIS imagery fails to detect even large rivers during certain periods clearly shows its limitations for detecting floods. Floods must spread over a large share of the 250 m pixel

Figure A12: Progression in water detection across MODIS and Landsat imagery.



Note: Data are from the 250 m MODIS MOD09GA (top) and 30 m Landsat LE07/C02/T1.L2 (bottom) collections on Google Earth Engine. All data are from 2012. We present three cases with rivers of different widths to illustrate how the ability of MODIS to detect surface water depends on the spatial scale of the underlying feature. For each case, we visualize median NDWI during the January-July non-flood period and maximum NDWI during the August-November flood period for both MODIS and Landsat. NDWI values above 0 are more likely to indicate the presence of surface water.

area to be detectable using NDWI-based algorithms. Floods that spread unevenly over space – likely in uneven terrain – are unlikely to be detected. MODIS imagery is therefore likely capture only a subset of more severe floods or floods in particular geographic conditions, which can further help explain the large number of GHSP communities with 2012 floods not identified in the NASA database.

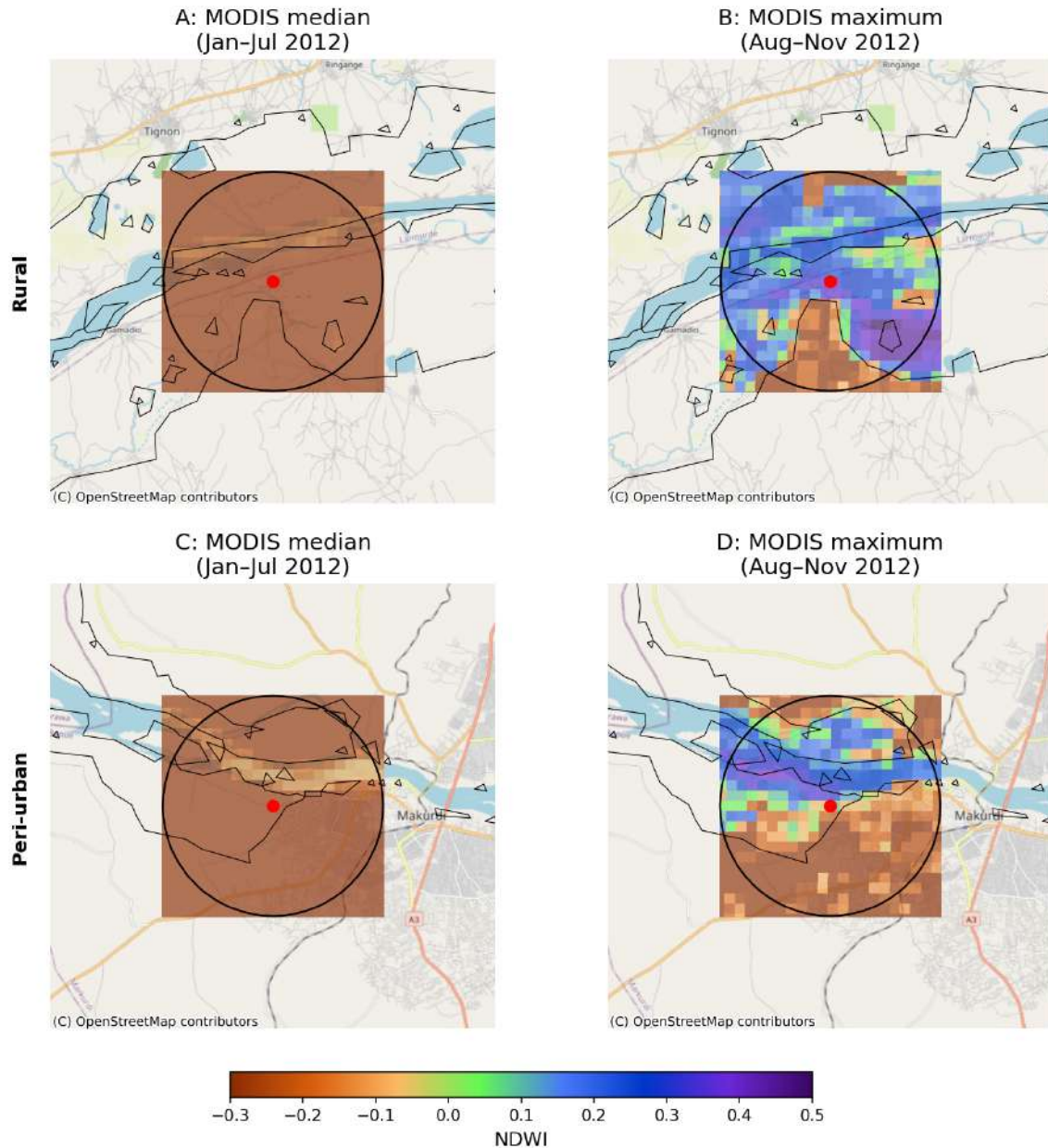
D.3 Illustrating survey vs. satellite flood detection

To illustrate how survey-based and satellite-based flood measures can agree or disagree at the community level, we present visual evidence from selected community examples. These examples are not intended to validate one source against the other mechanically. Rather, they show how the spatial distribution of water, uncertainty about the true location of the community within the buffer, and the construction of the MODIS-based indicator help interpret the relationship between the two measures.

Figure A13 presents two cases of agreement between the survey and satellite flood measures we use, considering rural and peri-urban communities near rivers. In both cases, we find much higher NDWI values in MODIS data during the flooding period than during the non-flooding period, with clear alignment with the MODIS-based flood detection boundaries. The figures appear to show clear cases of riverine flooding in large parts of the 5 km buffer around the communities.

Figure A14 presents examples of disagreement between the two flood measures. Panel a) shows two communities where flooding is detected within 5 km of the community centroid by the NASA

Figure A13: Examples of survey-satellite agreement in rural and urban settings



Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. The left panels show median MODIS NDWI in the non-flood period (January–July 2012), while the right panels show maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA Global Flood Product layer used for the satellite-based flood measure.

GLobal Flood Product, but is not reported in any household or community surveys.²⁸ In the top row, changes in NDWI during the flooding season suggest a broader increase in local surface water in what appears to be a hydrologically complex setting characterized by marshes, so the satellite flood identification may reflect real water expansion that was not experienced as a flood.

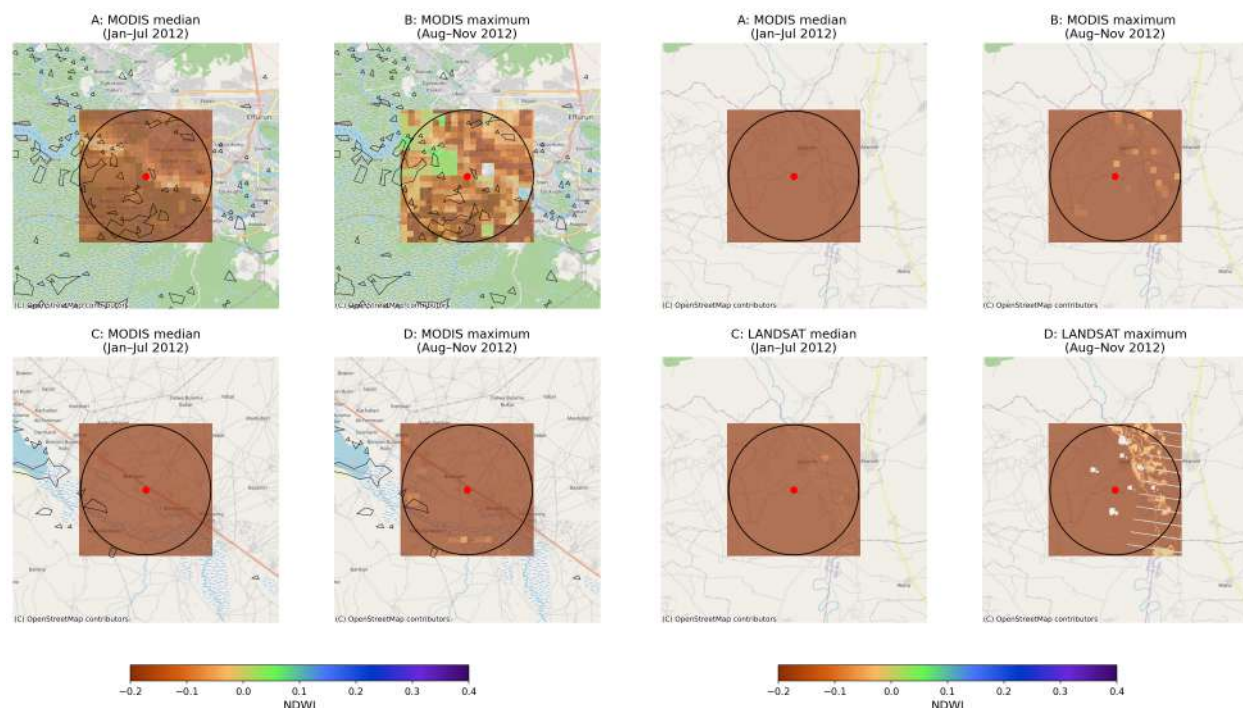
²⁸In 17 of 73 communities with satellite-only flooding, there is no flood according to the main survey-based measure requiring at least 2 household flood reports, but there is at least one household or community survey flood report.

In the bottom row, visible change in NDWI within the buffer is much weaker, yet the community is still classified as flooded because the NASA flood layer intersects the 5 km buffer. The mismatch may represent either a limitation of the buffer based matching process for the NASA flood layer – required due to uncertainty about exact community locations – or a limitation or error in the NASA flood layer itself.

Figure A14: Examples of survey-satellite disagreement

(a) Satellite-only cases

(b) Survey-only case



Note: The figure visualizes changes in NDWI and flood detection in a 5 km buffer around a rural GHSP community (top row) and an urban community (bottom). The jittered community centroids are indicated by red dots, and the 5 km buffers by black circles. For each pair of images, the left panel shows median MODIS NDWI in the non-flood period (January–July 2012), while the right panel shows maximum MODIS NDWI in the flood period (August–November 2012). NDWI values above 0 are more likely to indicate the presence of surface water. Black polygons denote the MODIS-based NASA Global Flood Product layer used for the satellite-based flood measure. Panel A presents two communities classified as flooded by the MODIS-based indicator but with no survey flood reports. Panel B presents a community with at least two survey flood reports no MODIS-based flooding detected within 5 km. For comparison, we also show changes in NDWI using higher-resolution Landsat which suggests a clearer pattern of riverbed expansion.

Panel b) illustrates the case of a community with multiple survey flood reports but no flood detection within 5 km in the NASA database. Changes in NDWI captured by MODIS suggest only weak differences during the flood season, but Landsat data suggests a broader pattern of localized riverbed expansion, some of which may have been experienced as flooding. This is the type of setting in which spatial resolution matters most: localized flooding may be meaningful for households while remaining only weakly visible in MODIS.